

C	131	P61: This could improve in future owing to low chances of employment	Maybe better in future
C	132	P62: Has been a permanent problem	Permanent problem
C	132	P63: Preyed upon my mind	Preyed upon my mind
C	132	P64: Psychologically no longer able to undergo repeat examination	Therefore no longer able to take repeat examination
C	132	P65: I won't manage a grade one	Not a grade one candidate
C	132	P66: Has worn down self-confidence	Self-confidence worn down
C	132	P67: Has never doubted own ideas of ability to deal with children	No self-doubts, however
C	132	P68: Emaciates, gnaws at one's own ego	Gnaws at one's own ego
C	133	P69: Some people who have more teaching ability are not bothered by this at all	Other people are less bothered
C	133	P70: People who do everything they are told	Conformists are less bothered
C	133	P71: May be too fine a point	May be too fine a point
C	133	P72: People who are more lively, more sociable, have new ideas and criticize in a witty manner ("master-of-ceremonies"-types) are very popular	"Master-of-ceremonies"-types are less affected
C	133	P73: Is, however, a question of mentality, cannot be made into a yardstick	Cannot be made into a criterion
D	134	P74: Had low pedagogical/ideological expectations myself	Had no preconceived ideas
D	134	P75: Hoped simply to do a good job	Hoped simply to do a good job
D	134	P76: Didn't work out nevertheless	Didn't work out nevertheless
D	134	P77: Had no practice	No practice

K9: Great practice shock because

- lack of practice
- seen by students as only a trainee
- criticism of seminary instructors destroys self-confidence and creates great pressure

K10: Only gradually learnt to deal with class without chaos

D	134	P78: Only accepted by the pupils as a teacher, not as a human being	Only accepted by pupils in the role of a teacher
D	134	P79: This is also due to the number of teacher trainees the children are exposed to	Too many teacher trainees
D	134	P80: Pressure from seminary instructors	Pressure from seminary instructors
D	134	P81: Do you down with criticism	Pressure through criticism
D	134	P82: More or less no self-confidence	Self-confidence destroyed
D	134	P83: Self-assuredness and authority thereby difficult to maintain in class	Stance in the class made difficult
D	134	P84: Insoluble conflict	Insoluble conflict
D	134	P85: Chaos in the class in seminary training school	Initially chaos
D	134	P86: Branch school better	Branch school better
D	134	P87: Knocked the stuffing out of me	Knocked the stuffing out of me
D	134	P88: Came out feeling very small	Self-confidence destroyed
D	135	P89: Positive experiences destroyed through criticism of seminary instructors	Positive experiences destroyed by seminary instructors
D	135	P90: You have the feeling that what you did was only a heap of trash	Self-confidence destroyed
D	135	P91: After a time got on well with the class after all	After a time got on well with the class
D	135	P92: This was not accepted by the seminary instructor	Not accepted by seminary instructor
D	135	P93: Chaos at the beginning	Chaos at the beginning
D	135	P94: Shock at seminary instructor	Shock at seminary instructor
D	135	P95: Shock at the boisterous classes	Chaos at the beginning

D	135	P96: Didn't manage to assert myself, quieten class down for lesson	Chaos at the beginning
D	135	P97: This entails use of a certain method which must be learnt	Getting on with the class is something one can learn

With the 10 categories of the right-hand column complete, we have now finished the first summary. In a second run-through these categories should be further reduced. In order to achieve this, the level of abstraction is raised. The statements are now intended to transcend the single case, no longer portraying the assessments of the individual teacher, but being generalized to an overall evaluation of the postgraduate training phase with its "practice shock". Certainly, such a generalization on the basis of just four case studies is not entirely justified content-wise, but it will nevertheless be carried out here for purposes of demonstration.

Case	Category	Generalization	Reduction
A	K1: Practical teaching not experienced as a shock, but as very enjoyable, because - previous teaching experience; - country school without discipline problems; - had no unrealistic expectations; - had good relations to students	No practice shock if: - previous teaching experience - good conditions - no unrealistic expectations Good relations to students possible	K'1: No practice shock occurs, if one - has had prior teaching experience; - has favorable training conditions in the postgraduate phase; - is flexible and adaptable; - communicates openly with colleagues; - has no "unrealistic" pedagogical expectations (illusion of simple persuasion techniques).
A	K2: Without these conditions practice shock undoubtedly conceivable	Otherwise practice shock	
B	K3: No practice shock, owing to flexibility, realistic attitude, adaptability and conversations with open colleagues	No practice shock if - flexible and adaptable; - conversations with colleagues	

B	K4: Belief in getting by without disciplinary measures, just on the strength of persuasion an illusion, because - even experienced teachers have difficulties; - students expect disciplinary measures; - large classes; - frequent change of class; - relativity of educational values; - good relation to students is also possible on a different basis	No practice shock if illusion of being able to get by without disciplinary measures is given up Good relations to students possible	K'2: Practice shock can reduce and strain self-confidence considerably, if - no practice was experienced beforehand; - destructive criticism and obligation to adapt to seminary instructor are not "taken in stride"; - one is not completely convinced of oneself K'3: A good relationship with students can always be attained K'4: Wanting to try out pedagogical behavior strategies and still remaining consistent in one's treatment of the class presents a dilemma
B	K5: Ski trips/sport/games can compensate for harsh image	Harsh image can be compensated for	
B	K6: Dilemma of trying out various pedagogical behavior strategies and nevertheless remaining consistent	Dilemma of trying out various pedagogical behavior strategies and nevertheless remaining consistent	
C	K7: Practice shock a great problem owing to obligation to adapt to ideas of seminary instructors in order to acquire good grades; gnawed at self-confidence, own ego	Being forced to adapt to seminary instructor can damage self-confidence	
C	K8: Perhaps due to - greater sensitivity; - not a grade-one candidate; - not a "conferencier" type; - less adaptable	Self-confidence in danger, - if more sensitive; - if not completely convinced of oneself; - if less adaptable	
D	K9: Great practice shock because - lack of practice; - seen by students as only as trainee; criticism of seminary instructors destroys self-confidence and creates great pressure	Practice shock, if - lack of practice; - lack of reputation among students; - destructive criticism by seminary instructor	
D	K10: Only gradually learnt to deal with class without chaos	Dealing with class can be learned	

The re-testing of the categories by applying them to the base material showed itself to be fairly representative. The purpose of summarizing qualitative content analysis is thereby fulfilled: viz., to reduce a large volume of material to a manageable level, but in so doing retaining the essential content. This reduction process can also be portrayed quantitatively; the breadth of the rectangles in the following is intended to represent the volume of material.

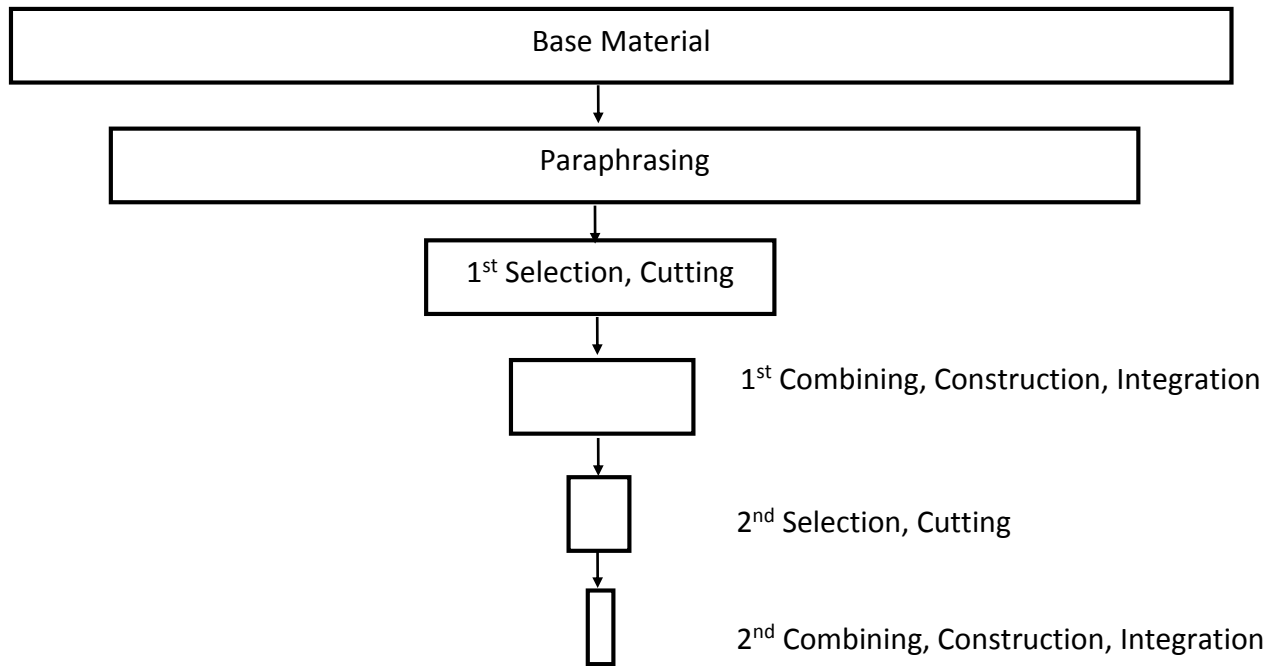


Figure 13: Material reduction through summary

6.3 Inductive Category Formation

Even if the reduction of the material with summarizing content analysis is impressive, the procedure is very extensive. Compiling those summarizing tables need nearly as much pages as the basic material. A second disadvantage of summarizing is that you have to consider all material, even if it is not relevant to the research question. Material for qualitative content analysis often stems from open-ended interviews, and those interviews sometimes wander away from the subject, what is tolerated because of a good relationship. Or the relevant content for the specific research question occurs at different points of the material.

So we developed a faster and more economic and more specific procedure in this context which we called inductive category assignment. The logic of summarizing, the theoretical background and plenty of rules are the same as summarizing content analysis, with three exceptions:

- Not all material is regarded for analysis. Only those parts relevant for a specific research questions are considered. For this selection process a rule of selection is formulated.
- The step of building paraphrases is skipped.
- The level of reduction is defined in advance, so that the category formulation can directly jump to this level.

So the aim is to arrive at summarizing categories directly, which are coming from the material itself, not from theoretical considerations. In so far the procedure can be called inductive category formation.

For qualitative content analysis this procedure is very fruitful. We have heard, that category definition is a central step in content analysis, a very sensitive process, "an art" (Krippendorff, 1980; cf. chapter 4). The inductive ongoing has great importance within qualitative research (cf. chapter 4). It aims at a true description without bias owing to the preconceptions of the researcher, an understanding of the material in terms of the material.

Inductive category formation is a central process within the approach of Grounded Theory (Strauss, 1987; Strauss & Corbin, 1990), which in this context is called "open coding". They developed a lot of rules of thumb for open coding; they recommended a systematic, line by line procedure. For content analysis, nevertheless, inductive category formation has to be more systematic. And it can use the same logic, the same reductive procedures, as in summarizing content analysis. The following process model (fig. 17) will now be explained.

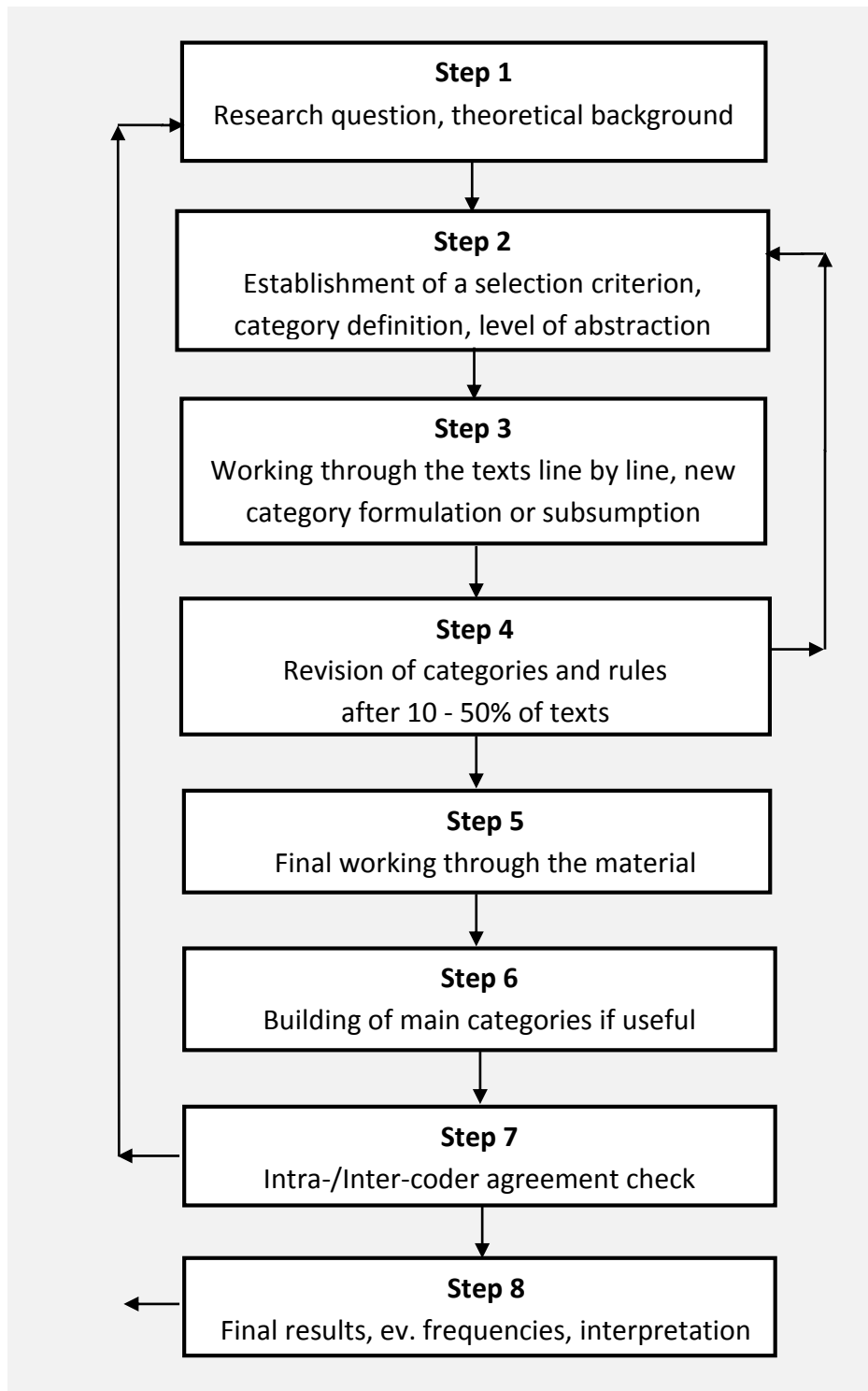


Figure 14: Steps of inductive category development

Within the logic of content analysis, the level or theme of categories to be developed must be defined previously. There has to be a criterion for the selection process in category formation. This

is a deductive element and is established within theoretical considerations about the subject matter and the aims of analysis.

After this is decided, the material is worked through line by line. The first time, material fitting the category definition is found, a category has to be constructed. A term or short sentence, which characterizes the material as near as possible (e.g. formulations if possible out of the material) serves as category label.

The next time a passage fitting the category definition is found it has to be checked, whether it falls under the previous category, then it can be subsumed under this category (a reductive process); if not a new category has to be formulated.

After working through a good deal of material (ca. 10 - 50 %) no new categories are to be found. This is the moment for a revision of the whole category system. It has to be checked, if the logic of categories is clear (e.g. no overlaps) and if the level of abstraction is adequate to the subject matter and aims of analysis. Perhaps the category definition has to be changed.

If there are any changes in the category system, of course the complete material once again has to be worked through.

Usually the level of abstraction is defined in a manner that fits best to the research question, and this is tested within the pilot phase (step 4). If too many categories had been formulated so that a clear picture of the object area does not occur, the level of abstraction should be defined more general. As a rule of thumbs, a set of ten to thirty categories gives a good overview. But sometimes it would be interesting to bring this set of categories into an order by formulation main categories. This step could be processed more inductively by only enhancing the level of abstraction in the sense of summarizing. It could be processed more deductively by introducing theoretical considerations in formulation main categories.

Link to QCMap software (www.qcmap.org):

On the project page in QCMap, a link to "Analysis" can be seen. Clicking on this link you can formulate main categories and subsume the inductive categories to those new main categories.

After this analysis we have a set of categories to a specific topic, connected with specific passages in the material. The further analysis can go different ways:

- The whole system of categories can be interpreted in terms of aims of analysis and used theories.

- The links between categories and passages in the material can be analyzed quantitatively. E.g. we can have a look at those categories occurring most frequently in the material.

Link to QCMap software (www.qcmap.org):

For inductive category formation the software offers three outputs (a link on the project page named "Analysis") as Excel-files: A list of all coded text passages, a list of all categories with frequencies and percentages (for an example see Table 5), and a table of categories per cases.

So the procedure rules for the single steps of inductive category formation (= I), based on summarizing, (cf. chapter 6.2) are the following:

I1: Research question

- I1.1 Formulate a clear research question (not only a topic)!
- I1.2 Describe the theoretical background (theoretical position, previous studies)!
- I1.3 The research question must fit an inductive logic, that means it must be explorative or descriptive in its nature.

I2: Category definition and level of abstraction

- I2.1 The category definition serves as selection criterion to determine the relevant material from the texts; it has to be an explicit definition, theoretical references can be useful.
- I2.2 The level of abstraction defines, how specific or general the categories have to be formulated. Both rules (category definition and level of abstraction) are central for inductive category formation. They have to be defined in advance and can be altered within the pilot phase.

I3: Coding the text

- I3.1 Read the material from the beginning, line by line, and check if material occurs that is related to the category definition! All other material is ignored within this procedure.
- I3.2 Formulate a category near to the text at the level of abstraction!
- I3.3 If the next passage fits the category definition, check if it can be subsumed to the first category or if a new category has to be formulated, and so on!

I4: Revision

- I4.1 A revision in the sense of a pilot loop is necessary, when the category system seems to become stable (only few new categories).
- I4.2 Check if the category system fits the research question! If not, a revision of the category definition would be necessary.

- 14.3 Check if the degree of generalization is sufficient! If you have formulated only few categories, maybe the level of abstraction is too general. If you have formulated a huge amount of categories maybe the level of abstraction is too specific.
- 14.4 If you have changed the category definition and/or the level of abstraction, you have to start the analysis from the beginning of the material!

15: Final coding

- 15.1 The whole material has to be worked through with the same rules (category definition and level of abstraction).

16: Main categories

- 16.1 At the end of this process you have a list of categories. You can group them and build main categories, if useful for answering the research question.
- 16.2 Follow the rules of summarizing qualitative content analysis (see book chapter 6.2) for this step!

17: Intra-/intercoder check

- 17.1 Start coding from the beginning of the material and compare the results (intra-coder agreement) (see book chapter 7 for this step)!
- 17.2 Give the material (or parts of it) to a second coder and compare the results. If the explorative character of the study is predominant, give him or her only the text. If the frequency distribution of the categories should be tested, give him or her your categories as well.
- 17.3 You should discuss the results and decide which coding is adequate (following the rules). Only if the second coding is held as better coding, this is counted as disagreement.
- 17.4 If you change the better coding for analysis you can enhance reliability (not always possible).

18: Results

- 18.1 The result (of course after checking quality criteria like inter-coder agreement) is at first the list of categories and maybe main categories.
- 18.2 If categories had been found in respect to several text passages (many subsumptions) a frequency analysis of the category occurrences could be useful.
- 18.3 The category system and eventually the frequencies have to be interpreted in the direction of the research question.

Example (resuming the project from chapter 5)

There is a distinct research question related to the interviews (appendix) which would allow a more economic procedure of text analysis taking into account only those text passages which relate to

the research question (in contrary to summarizing content analysis which has to consider all material):

Description of stress factors in first praxis experiences: First professional experiences, especially for teachers, are often described as "praxis shock" (Smagorinsky et al. 2004; Mueller-Forbrodt, 1978). We want to describe the concrete stressing factors.

Because the scope of analysis is more explorative we do not have a preformulated set of categories. This is a case for inductive category development.

We define the content-analytical units:

Coding unit: Clear semantic elements in the text

Context unit: The whole interview, interviewer protocol and background material

Recording unit: All four interviews (A to D)

The category definition is formulated as: Stressful experiences in and around teaching, experiences of harm, loss or challenge which are not automatically coped with (Lazarus).

The level of abstraction is: Concrete stress factors for the person, connected with negative experiences, no general evaluations of the situation, in a form that can occur as well in other interviews (no idiosyncratic formulations).

These are the codings and the text passages:

B1: Disappointments about students

"Certainly, there are disappointments that the students are not as one thinks they ought to be." (Case A)

B2: Little time for education

"what comes out at the end is very little, because 45 minutes, 35 students, that means for each student I've got a time ration" of about one minute." (Case A)

B3: Difficult students

"with the class I had initially, eighth graders, a bit of a difficult lot" (Case B)

"I've got problems with this or that student" (Case B)

B4: Problems in very large classes

"I always had very large classes, you see, in geography above all, 30 was the smallest number, but I often had around 38, and that's, I mean, then there really are situations arising which cause problems" (Case B)

B5: Being forced to authoritarian behaviour

“you're really forced then to do things, act in a way that - (laughs) to be really honest - I could never have imagined”

B6: Dependence on seminar instructor

“dependence on the seminary instructors” (Case C)

“somehow looking to get assessments which were as good as possible” (Case C)

“you have to fit in from the beginning with what the seminary instructor has in the way of ideas and policies” (Case C)

“the pressure from seminary instructors... That you, they make you feel so small, everything - every word, every gesture, everything. Whoever you are, they'll first destroy you through criticism. All they do is criticize, that was the case with me.” (Case D)

B7: Conflicts with concepts different to the ones in mind of the seminar instructor

“A plan or an idea of how he can best fulfil the expectations of the seminary instructor and that of course leads to a conflict situation” (Case C)

B8: Forced by seminar instructor to apply mechanical rules

“Oh yes, it was, because I didn't, because I'm not really the type that can apply mechanical rules” (Case C)

“How can that sort of thing be assessed (laughs) or made into a yardstick?” (Case C)

B9: Critique by seminar instructor impacts negatively on self-esteem

“it eats away at you, and for that reason - makes inroads into your self-esteem” (Case C)

“they make you feel so small, everything - every word, every gesture, everything. Whoever you are, they'll first destroy you through criticism. All they do is criticize, that was the case with me. And then you are, your self-confidence is zero-level” (Case D)

“And you thought, my God, what am I? Your self-confidence..that all you'd done the whole year was apparently nothing but rubbish, that nothing you'd ever done was correct. That's the feeling you have.” (Case D)

B10: Lack of experiences in teaching

“No! (laughs). It didn't work. I mean, let's put it this way: these pragmatic demands, expectations, they're in any case a bit, they're rather petty, unimportant, not even they worked. And the reasons were a) because one has had no experience” (Case D)

B11: Inferior teacher role as trainee

“The children in a seminary school like that always say, aha, here comes another new teacher trainee.” (Case D)

B12: To calm down an agitated classroom when left alone without seminar instructor

“When we were left alone the first time, without a seminary instructor sitting at the back, they went mad, all hell was let loose and (laughs) that was the first shock, how to go about imposing your will on the class for the first time.” (Case D)

So we arrived at twelve inductive categories which can describe very well the stress situation of the teacher students.

Some text passages at first glance seem relevant for coding; but a further look at the content-analytical rules excludes them. This is the case in the middle of the first interview (Case A):

“If you're at a school, for instance an inner city school where you've got discipline problems, where the students just – are completely different personality-wise, then maybe you do get somehow frustrated as a teacher. But in my case...”

The person is speaking about stress factors, but not for himself, and this was part of the category definition, so no coding is made.

Other text passages indicate stress for the person, but the formulation is too general, unspecific and so could not be coded (Level of abstraction!), for example in case B:

“and if anything shocking really happened, then –”

If you have coded more material, more interviews, then a frequency analysis of the coded inductive categories can make sense. It would be interesting, which categories occur most frequently and so represent the most imminent stress factors. A next step could be to compare the most frequent categories between different groups of persons (e.g. female and male). Crosstabs could be calculated and tested if certain persons show significant differences in the occurrence of certain categories. For example we could ask whether the category B6 (Dependence of the seminar instructor) is mentioned more often by younger teacher students.

The results can be displayed in a table, ordering the categories following the frequency of their occurrences in the material. Two aspects on category frequencies would be interesting: the absolute number of category occurrences within the material, and the number of different texts or persons (in the case of interviews, in our example: 4 persons) in which the categories had been coded. The frequencies can be displayed in absolute numbers and in percentages. For our short example the resulting table would be like this (Table 5):

Table 5: Category frequencies within the example

Category	N of C	% of C	N of P	% of P
B6: Dependence on seminar instructor	4	21%	2	50%
B9: Critique by seminar instructor impacts negatively on self-esteem	3	16%	2	50%
B3: Difficult students	2	11%	1	25%
B8: Forced by seminar instructor to apply mechanical rules	2	11%	1	25%

B1: Disappointments about students	1	5%	1	25%
B2: Little time for education	1	5%	1	25%
B4: Problems in very large classes	1	5%	1	25%
B5: Being forced to authoritarian behaviour	1	5%	1	25%
B7: Conflicts with concepts different to those of the seminar instructor	1	5%	1	25%
B10: Lack of experiences in teaching	1	5%	1	25%
B11: Inferior teacher role as trainee	1	5%	1	25%
B12: To calm down an agitated classroom when left alone	1	5%	1	25%
Σ	19	100%	4	--

Note: Row 1 categories ordered by frequencies; row 2 number of occurrences, row 3 percentage of all codings; row 4 number of persons; row 5 percentage of all persons

This gives a good overview of the different problems experienced by the teacher students. Most interesting for interpretation would be those categories with many occurrences (the first four in table 5). It would be legitimate to display a table only with those categories occurring in several text passages, to formulate a cut-off criterion.

To formulate main categories within the list of twelve categories could make sense. In this example a more inductive way could lead to three main categories:

B'1: Problems in relationship to students (B1, B3, B5, B10, B12)

B'2: Structural conditions (B2, B4, B11)

B'3: Seminar instructor (B6, B7, B8, B9)

6.4 Explication (Context Analysis)

Whereas the goal of summarizing content analysis and inductive category formation was the reduction of the material, the tendency of explication is exactly the reverse. Individual parts of text in need of interpretation are enriched by additional material aimed at explaining them, making them comprehensible, subjecting them to comment and illustration.

The basic idea behind explication as a qualitative content-analytical method is that it precisely defines which additional material is permissible to explain a certain point in the text. For the quality of the interpretation depends on the material chosen.

Every interpretation must have as its basis a lexical-grammatical definition; the meaning of language, within its cultural context and in its respective current forms, is continually portrayed in dictionaries and other works of reference; sentence structures are determined in grammars. Knowledge of this general lexical-grammatical character of the particular point of the text concerned is the precondition for the interpretation of it.

However, the analysis takes on a particular interest and importance when the speaker deviates from this general usage and starts conferring on language items his own specific personal meanings, or expresses himself in an unclear or incomplete manner. In this case, the analyst must resort to the context in which the utterance occurs. Techniques of explication vary according to how broadly this context is defined.

Thus Volmert (1979) differentiates on this point between spatially restricted textual emphasis (i.e. the direct references in the text), and spatially extensive emphasis (which takes account of factors such as information already given, background knowledge, the horizon of comprehension, but equally the behavioral context, the non-verbal context and the situational context of the portion of text to be interpreted). Van Dijk (1999; 2007) has introduced the concept of micro context and macro context (see chapter 3.3).

In this connection we shall distinguish here between a narrow and a broad contextual analysis. The interpretation objective must then be, on the basis of the contextual analysis, to arrive at a statement so phrased that it constitutes a key to understanding the portion of text in question. It can then be established within the total context of the material whether this explication is sufficient or not. On the basis of these considerations we will now formulate and explain a general procedural model of interpretation (Fig. 15).

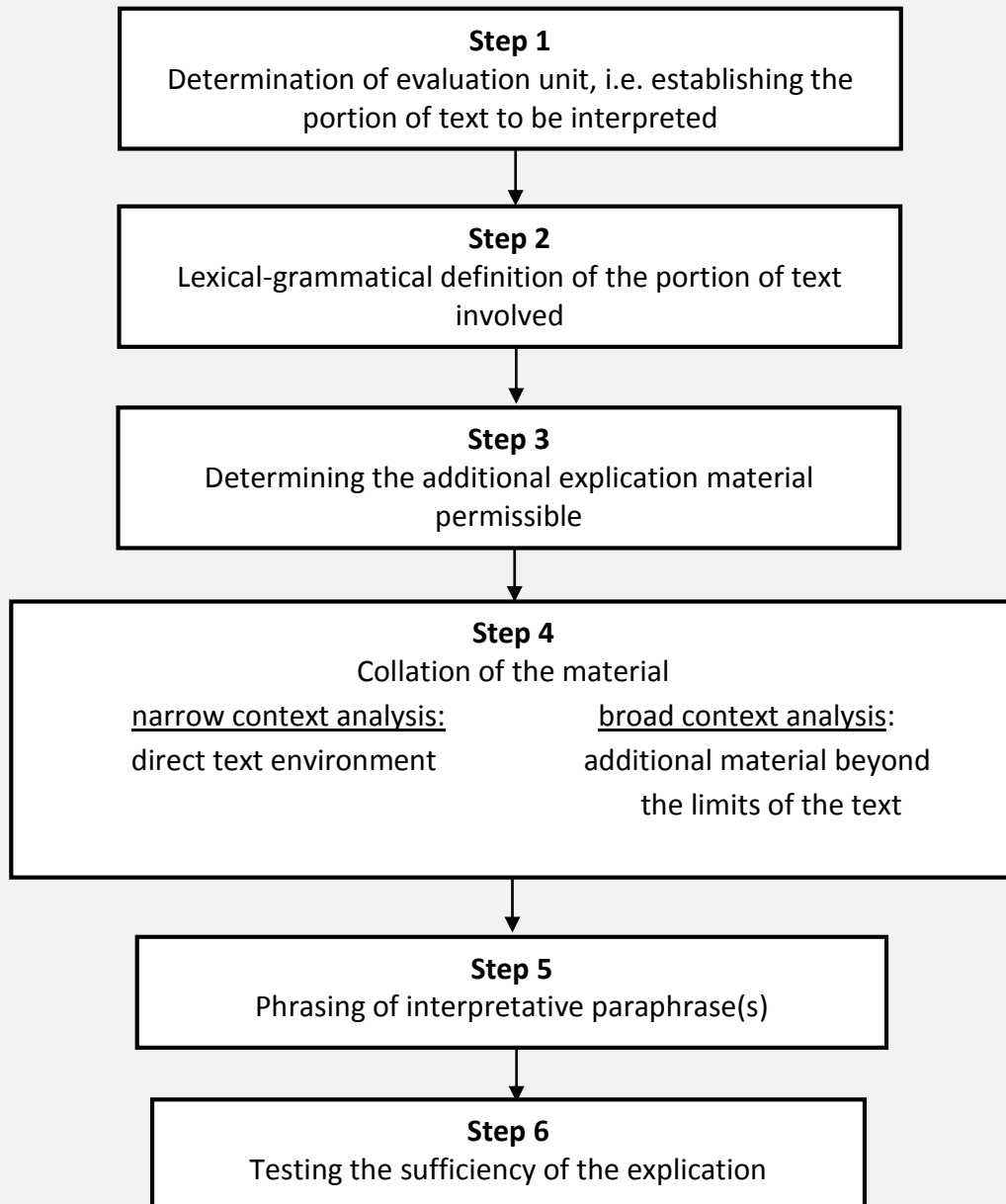


Figure 15: Procedural model of explicational content analysis

The starting point of explication is the exact definition of the portion of text to be interpreted (Step 1). The definition depicts the evaluation unit of the analysis. The determination of the encoding unit coincides here with the contextual unit, as what is to be used as context material is encoded during the explication. This does not occur, however, until later in the analysis.

The second step examines whether the portion of text can be interpreted through grammatical analysis or on the basis of lexical meaning alone. In this connection it is important to consider which grammars and reference dictionaries of the respective linguistic and socio-cultural environment are relevant to the task. The translation of a text or passage, which in the widest sense could also be understood as explicational content analysis, would already be completed during this stage of the proceedings.

As a rule, however, this is not sufficient for the proper explication. Thus in the third step it must be determined what additional material is to be allowed for the interpretation. The rule here is that one proceeds from the narrowest context to successively broader ones.

During the collating of material that now follows (Step 4) a distinction must be drawn between narrow and broad contextual analysis.

Narrow contextual analysis admits only material taken from the text itself. Passages which are directly related to the particular passage in question are collected from the whole text.

Such passages can stand in

- defining, explanatory,
- embellishing, descriptive
- exemplifying, itemizing,
- correctional, modifying,
- antithetical or contradictory

relationship to the passage in question.

In addition, the narrow context analysis examines whether the passage to be explained occurs in similar or identical form elsewhere in the material. If so, the narrow textual context at that point is also included for analysis. Material going beyond the actual text is then collected for the broad context analysis. Such material may include information on the author of the text (cf. point 4.6, Definition of base material), or information on the conditions of origin of the text (cf. point 4.6). But interpretatory material may also be derived from preliminary theoretical conceptions (cf. point 4.6, Theory-bound differentiation of the issue). The broadest form of context analysis permits use of the entire background understanding of the analyst(s) in the interpretation. This can go as far even as the analyst's using free association on the contents in the passage concerned (cf. the second example of a qualitative analysis of biographical documents in Gstettner, 1980). In the case of such explication material, certainly, its relevance and relation to the text passage must be justified precisely.

The next step (Step 5) then consists of constructing a statement which explains the passage in question. An explicative paraphrase of this kind usually comes about through the summarizing of the collected material (cf. the rules of summary). If inconsistencies occur in the material, however, it is necessary to formulate alternative paraphrases.

In the last stage (Step 6) the paraphrase (or the alternative paraphrases) is positioned in the text at the place of the passage to be interpreted, to test in the overall context whether a sensible explication has been attained. If this is not the case, new explication material must be decided upon and a new run-through of the context analysis carried out.

From this description of the procedural model we can now draw up interpretation rules for explicating content analysis:

E1: Lexical-grammatical definition

- E1.1 Determine the dictionaries and grammars relevant to the linguistic and socio-cultural background.
- E1.2 Then analyze the lexical and grammatical meaning of the passage.
- E1.3 Examine whether this already explains the passage adequately.

E2: Determination of the explication material

- E2.1 Begin with the narrowest textual context, i.e. with the immediate environment of the passage in the text which has to be explained.
- E2.2 Proceed to successively broader contexts if the check on the explication was not satisfactory.

E3: Narrow context analysis

- E3.1 Collate all the statements in the immediate textual context which are directly related to the passage in question, i.e. in a
 - defining, explanatory,
 - embellishing, descriptive
 - exemplifying, itemizing,
 - correctional, modifying,
 - antithetical or contradictory manner.
- E3.2 Check whether the passage to be explained occurs elsewhere in identical or similar form and if so examine the immediate textual environment of the places where it occurs.

E4: Broad context analysis

- E4.1 Check whether further explanatory material is available on the author of the passage.
- E4.2 Include material on the situation of the origin of the text in the explanatory process.
- E4.3 Check whether explicational material can be derived from preliminary theoretical considerations.

E4.4 On the basis of your own general background of understanding check whether further material should be included or not.

E4.5 Explain the relevance, the relation of the material collected to the passage in question.

E5: Explicational paraphrase

E5.1 Summarize the material gathered for explication (cf. summary) and formulate from it a paraphrase for the passage in question.

E5.2 If the material is inconsistent or contradictory formulate several alternative paraphrases.

E6: Checking the explication

E6.1 Insert the explicatory paraphrase in the material in place of the passage in question.

E6.2 Check whether, in the overall context of the material, the passage is now appropriately expressed.

E6.3 If the explication does not appear adequate, decide on new explication material and run through the analysis again (from Step 3).

This will now be demonstrated using the example.

Example

In our sample material there is a passage which even in the summary appeared rather unclear. This is where Case C (see page 133 in the appendix) reports that he is not a "master-of-ceremonies" type and therefore somehow had a harder time during postgraduate training. This conception of the "master-of-ceremonies-type", the meaning of which appears at first sight rather obscure, will now be used to initiate an explicational content analysis.

Step 1: The passage to be explained is clearly marked: the problem revolves around the term "master-of-ceremonies-type" on page 8.

Step 2: In order to determine the lexical meaning it is necessary to consult relevant works of reference, i.e. modern dictionaries of Standard English [in the original: "of High German", trans. note]. The entry under "master-of-ceremonies" [in German "Conferencier", trans. note] lists, for instance, the following definitions: "Announcer on a small variety stage" (dtv-dictionary, vol. 3, 1966, p. 168) or "(witty and entertaining) announcer in cabaret, variety, at public and private functions" (Meyers Grosses Taschenbuchlexikon, vol. 5, 1981, p. 5).

However, such definitions do not help us very much to understand the term in the material context.

Step 3/Step 4: For the determination of permissible additional material we can refer first of all to the direct textual environment. The phrase within which the term was used is:

"I'd say it's very important, especially in sport, and I'm certainly not the type, not at all, no - well, I wouldn't quite say extrovert, but the more lively you are personally, in speaking or dealing actively with adults, or constantly - having new ideas or even making the odd criticism of seminary instructors, but in a witty or jocular way, more a "master-of-ceremonies" type; they are a great success, I believe. ... But that of course is a question of mentality. How can that sort of thing be assessed (laughs) or made into a yardstick?"

(Case C, p. 133)

The descriptive features mentioned here are:

- extrovert (?);
- lively when they speak;
- lively way of associating with adults;
- always having new ideas;
- express criticism of seminary instructor, phrased as a joke or witticism.

So one could say that a "master-of-ceremonies-type" is an extroverted, lively, witty person.

A further passage also seems to relate to this concept, which occurs in the script shortly beforehand: "Although it varies according to what type you are, I think. Some are not so bothered, they put on more of a face, they regard it more as, let's say you could see it this way, that the educational qualities they already have, though I'd put "educational qualities" in inverted commas, that they say to themselves, well, it has to be done like that, it has to be done like that, and then they do it like that. And if they're lucky it goes well for them, precisely because they've done it like that, and that's all right, isn't it." (Case C, p. 133)

Although the statement is a little confused, new descriptive features start to emerge:

- plays more;
- seems to bring the "pedagogical" abilities with him;
- always knows what is to be done;
- behaves accordingly always;
- is assessed well because of that.

The first statement about "playing" seems particularly important to me, although it is not enlarged on any further. This may explain the negative undertone of the remark about what essentially are very positive personality features. By "playing" the speaker probably means something along the lines of "playing a role", "having a trick up one's sleeve" to help one manipulate the situation to one's best advantage, thus in essence being "dishonest", i.e. simply play-acting.

This meaning also tends to correspond more to the lexical meaning, for a master-of-ceremonies is connected with acting in the theatre.

The remarks following from this second passage all tend in the direction of a person convinced of himself and his own worth.

Step 5: If these personal features are summarized in explanatory form, what we have on the one hand are:

- extrovert
- lively
- witty
- self-confident

and on the other: the feature "acting a part". Thus we can say that a master-of-ceremonies is someone who plays the role of an extrovert, lively, witty, and self-confident person.

Step 6: For purposes of checking, this interpretation must be placed in the context of the material. The context is to be found shortly before the place first quoted (p. 133) and shortly after the second place (p. 133).

- The master-of-ceremonies-type is not bothered so much by stress caused through pressure to adapt and blows to self-confidence.
- The MOC-type is more popular with seminary examiners.
- Being a MOC-type is a question of mentality.
- It is unfair to regard a mentality feature of this kind as a factor in assessment, as a yardstick for measuring pedagogical abilities.

If the paraphrase formulated in Step 5 is now inserted into these remarks, the result is a clearly comprehensible statement with an unambiguous meaning.

This explicational content analysis is now complete. Certainly, it would be possible to collect further material on the speaker from the interview as a whole, concerning, for instance, the description of his teaching practice and his examination experiences. In this case a new run-through would have to be done. But this does not appear to be necessary.

And so we will now pass on to the description of the next qualitative technique, that of structuring content analysis.

6.5 Structuring – Deductive Category Assignment

This is the content-analytical method which is probably most central. It has the goal of extracting a certain structure from the material. This structure is brought to bear on the material in the form of a category system. All text components addressed by the categories are then extracted from the material systematically. If one wishes to describe the structuring procedure quite generally, a few points, it seems to me, are especially important. The fundamental structuring dimensions must be exactly determined. They must derive from the issue/statement of the problem concerned, and must be theoretically based. These structuring dimensions are then, as a rule, further subdivided, being resolved or split up into individual features or values. Subsequently, the dimensions and values are brought together to form a category system.

The particular categorization of a given material component is something that must be determined precisely. A procedure for this has proven useful (cf. Ulich, Hausser, Mayring, Strehmel, Kandler, Degenhardt, 1985; Hausser, Mayring & Strehmel, 1982). It can be justified by the approach of multiple systems in the categorization theory (see chapter 3.5). We have shown in chapter 3.5 that the theories of categorization from General Psychology could be the basis for this process, which operates in three stages:

1. Definition of the Categories

It is precisely determined which text components belong in a given category.

2. Anchor samples

Concrete passages belonging in particular categories are cited as typical examples to illustrate the character of those categories.

3. Coding rules

Where there are problems of delineation between categories, rules are formulated for the purpose of unambiguous assignment to a particular category.

Test extracts are taken from the material to check whether the categories are at all applicable and whether the definitions, anchor samples and encoding rules make categorical assignment possible.

This trial run-through, like the proper main run-through, is sub-divided into two steps of operation. First of all the text passages in the material are marked in which the category concerned is addressed. These "points of discovery" (cf. Hausser, Mayring & Strehmel, 1982) can be marked by noting the category number in the margin of the text or through differently colored underlining or marks in the text itself. In the second step the material thus marked is processed in accordance with the structuring intention (see below) and copied out of the text.

As a rule this trial run-through results in a revision and partial reformulation of the category system and its definitions.

Now the main material run-through can finally begin, again split up into the two stages of marking the points of discovery and extracting and processing them.

In accordance with the type of structuring (see below), the results of this run-through must then be summarized and analyzed.

This general description of a structuring content analysis can be shown in a procedural model as follows:

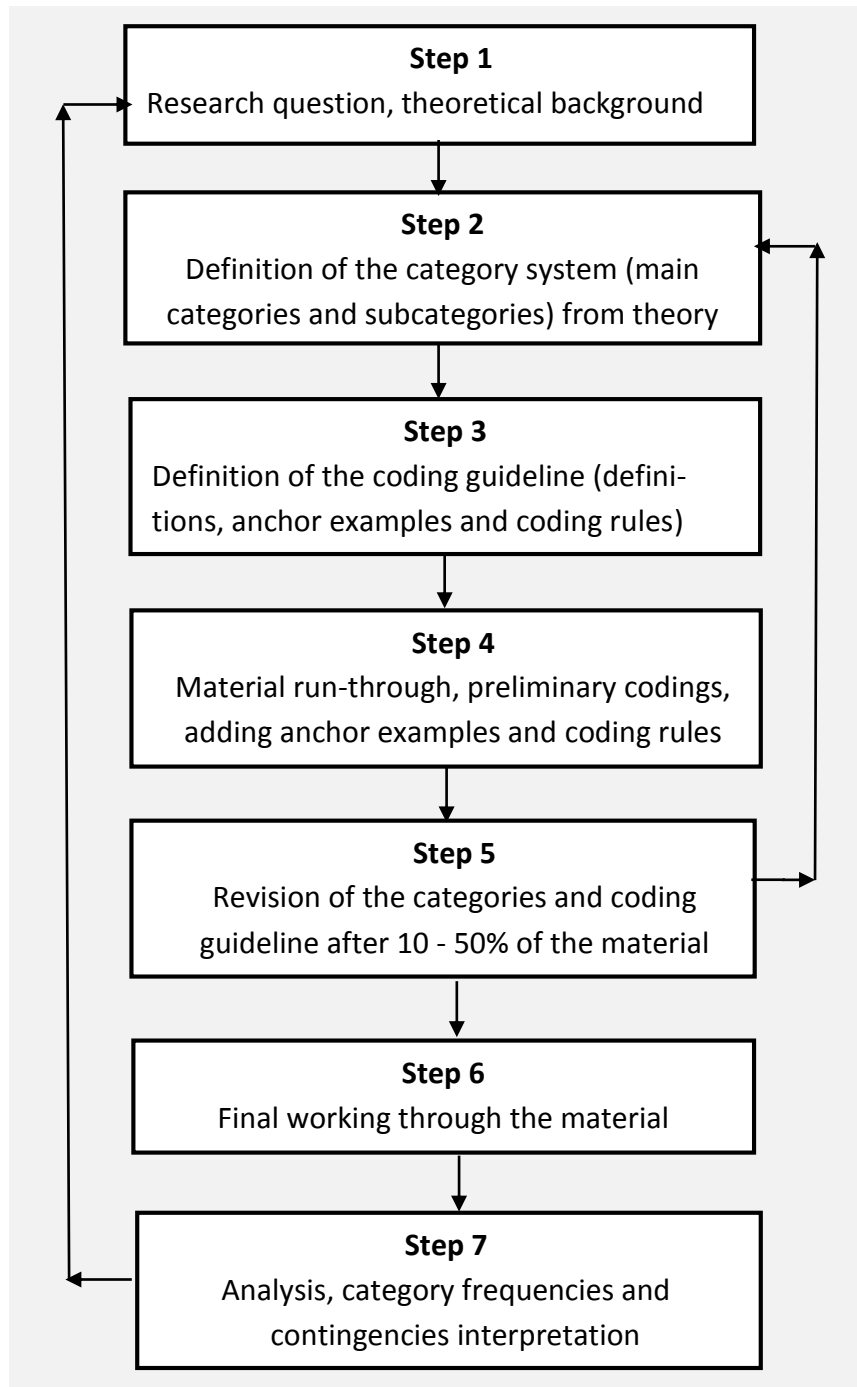


Figure 16: Steps of deductive category assignment

The procedure is deductive because the category system is established before coding the text. The categories are deduced from theory, from other studies, from previous research. Theoretical considerations can lead to a further categories or rephrasing of categories from previous studies, but the categories are not developed out of the text material like in inductive category formation.

So deductive category assignment is the adequate procedure if there is relevant previous research (less for explorative research designs, cf. chapter 1.5).

The procedure rules for the single steps of deductive category assignment (= D) are:

D1: Research question

- D1.1 Formulate a clear research question (not only a topic)!
- D1.2 Describe the theoretical background (theoretical position, previous studies)!
- D1.3 The research question must fit the deductive ongoing, that means that there is an a priori interest in special aspects of the material and a clear theoretical background.

D2: Definition of categories

- D2.1 The research question has to be operationalized into categories that means research aspects brought to the material.
- D2.2 Analyze the state of the art, preceding studies on the topic, to get a theoretical foundation! Not all categories have to be found in the research literature, but they have to be grounded with theoretical arguments!
- D2.3 Check, if the material contains text passages relevant to the categories!
- D2.4 If possible, try to group the categories to main categories in a nominal or ordinal way!

D3: Coding guideline

- D3.1 Formulate a table containing four columns: Category label, category definition, anchor example, coding rules! Each category represents one line.
- D3.2 Fill in the category labels and the category definitions, and, if already formulated, anchor examples and coding rules.

D4: Coding

- D4.1 Start coding the material from the beginning! If you find material fulfilling the category definition, mark the text passage and note the category label (or category number). If you think it is a prototypical text passage for the category, add it to the coding guideline as anchor example!
- D4.2 If you come to a text passage where the assignment to a category remains unclear, try to come to a decision and formulate a coding rule for this and following similar cases! In case of uncertainty use theoretical considerations!

D5: Revision

- D5.1 If the coding guideline seems to be completed (at least with anchor examples) and the coding process seems to be smooth (usually after 10 - 50% of the material) or if severe problems arise, a revision of categories and coding scheme is necessary!
- D5.2 Check all category definitions and coding rules in respect to the research question (face validity)!
- D5.3 If changes are necessary, use theoretical considerations!

D6: Final work through

- D6.1 If the changes of the coding guideline make prior category assignments false, you have to rework the material from the beginning!
- D6.2 List all category assignments linked to the recording units!

D7: Analysis

- D7.1 The result (of course after checking quality criteria like inter-coder agreement) is at first the distribution of categories per recording unit.
- D7.2 Frequencies of assigned categories over all recording units or comparisons of frequencies in different groups of recording units can be analyzed statistically.
- D7.3 In case of several ordinal category systems assigned to the same recording units, a correlation analysis (usually non-parametric) is possible.

There are two forms of deductive category assignment: analyzing the text with **nominal category systems** or with **ordinal category systems**. Nominal or qualitative category systems (cf. scales of measurement, e.g. Davis & Smith, 2005, p. 68 ff.) consist of a list of independent categories. The only similarity is that they are belonging to the structuring dimension. A list of fruits (C1: apples, C2: pears, C3: grapes, C4: lemons, C5: oranges ...) is a nominal category system. The difference to inductive category formation is that these categories are formulated in advance and hold constantly through the text analysis. The result looks similar: A list of categories related to text passages, eventually frequencies of their occurrences.

Ordinal category systems express a graduation of the structuring dimension. The categories are in a fixed order, following more or less the structuring dimension (e.g. K1: excellent, K2: good, K3: average, K4: bad). If we have assignments of ordinal categories to different units of analysis a broader range of statistical procedures can be used. For example, two ordinal category systems assigned to the same units of analysis allow the calculation of a (usually non-parametric) correlation coefficient.

Link to QCMap software (www.qcmap.org):

If you have uploaded textual material and formulated a research question in QCMap, you have to decide for the adequate content analytical technique. Choosing deductive category assignment opens automatically a new screen where you have to fill in the categories, definitions, anchor examples and coding rules. Only after this step you can code the texts.

This procedure of deductive category assignment (ordinal categories) will now be illustrated using the example text.

Example

Representing the central issues in the analysis of the sample material (cf. 5.2.9), two main questions were formulated, the second of which will now be dealt with using a structuring content analysis: Has the "practice shock" affected the self-confidence of the individual? Within the framework of the DFG project "Teacher Unemployment", from which the material is taken, this issue was examined for possible evidence of a generalized control expectancy on the part of the individual, which could also have an effect on the present situation (of unemployment) (cf. Ulich et al., 1985). With the operational procedure suggested here the attempt will be made to assess systematically and according to complex psychological variables biographical material compiled in retrospect. Whether this has been successful content-wise remains to be tested, as hitherto this is simply a first attempt. It can certainly serve well, however, as an example demonstrating the method of structuring content analysis.

Step 1: Determination of the units of analysis

When determining the unit of classification, the main question is when and how often in the material the evaluation (influence on self-confidence) is to be carried out. The first possibility is to designate the individual case as the unit of assessment. This, however, seems a little too rough.

If self-confidence is to be understood as the certainty of being able to cope well with demands of one's biographical development (cf. Step 2) then a good opportunity for the assessment of self-confidence presents itself if the latter is linked to such demands as they are portrayed in the material. This would provide a much more concrete unit of assessment: whenever demands on the individual are described as being initiated by the change from university to post-graduate training ("practice shock"), this is regarded as a unit of assessment.

The recording unit as the smallest text component which can fall within a category can now be determined as follows: as soon as the material within a unit of assessment allows the conclusion that the demand was coped with in a self-confident manner (definition of this in Steps 3 and 4), this can be encoded. In a purely formal sense it can even be a proposition as a minimal carrier of meaning.

As the context unit, finally, we have all the material that exists on the respective demand in a particular case.

Step 2: Establishing assessment dimension(s)

Self-confidence, a construct closely related to that of generalized control expectancy (Rotter 1966), will be inferred here from the way in which challenges are coped with in the individual's biography. Self-confidence is taken to mean the subjective certainty of being able to deal well with such challenges.

General self-confidence is therefore composed of individual, situationally specific values. This situation-specific self-confidence is the assessment dimension of our analysis. In order to infer self-confidence from the portrayal of a challenge in the material we have to define the concept more exactly. Self-confidence can be thought of as comprising a cognitive component, an emotional component, and an active component:

- being aware of the kind of challenge one is faced with and the strategies necessary to cope with it (cognitive component);
- having a positive, optimistic feeling in dealing with the challenge (emotional component);
- the certainty of being able to meet the challenge adequately (active component).

Step 3: Determining the values

As the material gives only rather scanty information on individual self-confidence we will use here a simple scale with three values on it: high - average - low. For all cases in which an unambiguous assignment to one of these three values is not possible, we will establish a reserve category: "not inferable". We therefore have the following categorization:

C1: high self-confidence

C3: low self-confidence

C2: average self-confidence

C4: self-confidence not inferable

Step 4: Definitions, anchor samples and encoding rules

The core of structuring content analysis, the exact description of the categories through definitions, anchor samples and encoding rules, which has been explained already in the general section, will now be demonstrated here in the form of an encoding guide. For the anchor samples, however, material from other scripts on the same subject and within the same project on "Teacher Unemployment" will also be used.

Table 6: Coding agenda for self-esteem

Variable	Value	Definition	Anchor samples	Encoding rules
Self-confidence	K1: high self-confidence	High subjective feeling of having met the challenge well, i.e. - good awareness of the kind of challenge and the way it should be coped with; - positive, optimistic feeling when dealing with the challenge - conviction that mastery of the challenge lay in one's own hand	"Of course there were little problems now and then, but they were simply solved: owing to a change either in my view or in that of the pupil, depending on who was at fault - we all make mistakes."	All three aspects of the definition must point in the direction of "high", at least no aspect should allow the diagnosis of simply average self-confidence; otherwise encoding for "average self-confidence"
Self-confidence	K2: average self-confidence	Only partial or fluctuating certainty of having coped with the challenge	"I managed to grope my way through O.K., but it was often a cliffhanger." "With time it got a bit better, but whether that had to do with me or with other circumstances I don't know." "Towards the end I got on quite well with the seminary instructor but I didn't have a very good feeling about it - I just accommodated myself, submitted to the demands."	If not all three aspects point to high or low self-confidence
Self-confidence	K3: Low self-confidence	Conviction of having coped badly with the challenge, i.e. - little awareness of the nature of the challenge; - negative, pessimistic feeling when dealing with the challenge; - conviction of not having had control of the way the challenge was dealt with.	"That hit my self-confidence hard, I thought of myself as a nobody, a nothing."	All three aspects point to low self-confidence, otherwise encoding for "average self-confidence"
Self-confidence	K4: self-confidence not inferable	The demands were reported but the manner of dealing with them remains unclear.	"At the beginning it was difficult, but with time it improved."	

Step 5: Marking of points of discovery

The marking of the text passages relevant to the categories, the first run-through of the material (if there are several run-throughs, with bigger amounts of material one text passage is sufficient), has to keep to the general definition of the categories (Step 1). Every point at which challenges posed by post-graduate training are mentioned in the material must be marked. Within such passages the specific portions of text allowing an evaluation of self-confidence should be underlined. In the sample text in the appendix of this book this is done by bold characters.

Step 6: Assignment of categories

Following the unit of analysis one of the four categories has to be assigned to each of the 4 cases. If there are several points of discovery within one case a comprehensive assignment has to be done. This is not a quantitative step (counting which category occurs most often within one case), but an interpretative act, following the coding agenda.

The individual codings with the arguments for the categorization are as follows:

Table 7: Deductive coding of example texts (appendix); t: top of page, m: middle of page, b: bottom of page

Case	Points of discovery	Code	Reasons for Code
A	p. 125 t p. 125 b p. 126 m	C1 (high)	Positive feeling (keen, enjoyment); explanation of disappointments (big city); conviction of mastery (looking forward)
B	p. 128 t p. 128 m p. 128 b	C1 (high)	Positive attitude; management of difficulties, always adequate reactions
C	p. 130 m p. 131 m p. 132 t p. 132 b	C3 (low)	Dependence on extern assessment, conflicts create problems (but perhaps over-sensitive?), erosion of self-esteem
D	p. 134 m p. 135 t p. 135 m	C2 (middle)	Problems because lack of experience, first destroyed by criticism, but awareness of the problems and mastery at the end

6.6. Mixed Procedures

As we have mentioned earlier there are there are possibilities to mix different basic procedures (inductive, deductive) in Qualitative Content Analysis. Depending on the research question, they offer interesting possibilities of text analysis. We will propose three possibilities (several others will be possible as well):

6.6.1 Content Structuring / Theme Analysis

In the first editions of “Qualitative Inhaltsanalyse” I proposed several forms of structuring (which we now call deductive category assignment); one of them was content structuring which meant to filter out from the material specific content dimensions and to summarize this material for each content dimension. If this is done inductively, the procedure is possible to implement by inductive category formation (cf. chapter 6.2). If the themes to be analyzed are fixed in advance (for example within an interview study the topics of the interview agenda), but the material per theme should be reduced, a combination of deductive and inductive procedures is needed.

Theme analysis or thematic analysis occurs in the content-analytical literature at several points. Stone (1997) defines it on the tradition of quantitative content analysis (Berelson, 1952) as selective analysis of subject matters or attributes of the text and formulates a bottom-up strategy (we would call it inductive) and a top-down (deductive) strategy. His aim is to identify themes as categories and to analyze frequencies and contingencies of the content categories. Boyatzis (1979) goes in a similar direction, describing thematic analysis as theory driven or data driven. Kuckartz (2014) conceptualizes thematic qualitative content analysis as a basically inductive process, Grounded Theory orientated.

In our context we only need to describe the more deductive sort of theme analysis, because inductive procedures are sufficiently described with inductive category formation. There are two basic steps of this form of content structuring or theme analysis:

- The first step is deductive. A list of themes is developed in advance, coming from theory, previous studies, from the interview agenda or sections of the data collection procedure. A coding guideline has to be developed, following deductive category assignment (cf. chapter 6.3). The material is coded with those categories.
- The second step is to extract all coded material per category and to summarize this material per category. If there is a huge amount of material per category, then inductive category formation is more adapted.

6.6.2 Type-building content analysis

In the first editions of “Qualitative Inhaltsanalyse” we have just described a form of type-building structuring. The label “structuring” is insufficient, because one central motive for finding typologies is to describe in deep those types (cf. Kluge, 2000). So this seems to be a mixed procedure.

The central idea of type-building is to classify and describe a heterogeneous amount of material. Typologies have a long tradition within social and behavioral sciences. The four temperaments (choleric, melancholic, sanguinic, phlegmatic) go back to antique thinking (Galen of Pergamon, 130 - 200p). Until the first half of 20th century, typologies were common in psychology as personality traits (e.g. C.G. Jung: introversion, extraversion). Max Weber developed the approach of ideal types for sociology. The Marienthal study of unemployment in the 1930ies (Jahoda, Lazarsfeld & Zeisel, 2002) has found four different reaction types: the unbroken, the resigned, the despaired, and the apathetic people.

Ritchie, Lewis, Nicholls & Ormstrong (2014) have worked out, that typologies can be simple descriptive, single-dimensional or more complex, multifactorial or multidimensional like a cross tabulation (Lazarsfeld & Barton, 1951). On the other hand, a different logic of type-building is possible. Are we looking for types as representatives of the most frequent occurrences within the chosen dimensions, or for extreme types (the typical best, the typical worst) or are we interested in certain values within the dimensions from a theoretical point of view? So the development of a typology needs different steps (cf. Kluge, 2000; Kuckartz, 2014): The definition of the dimensions within types and the logic of types should be formulated, the identification of types in the material and the description of those types. Within Qualitative Content Analysis this means the following steps (Figure 17):

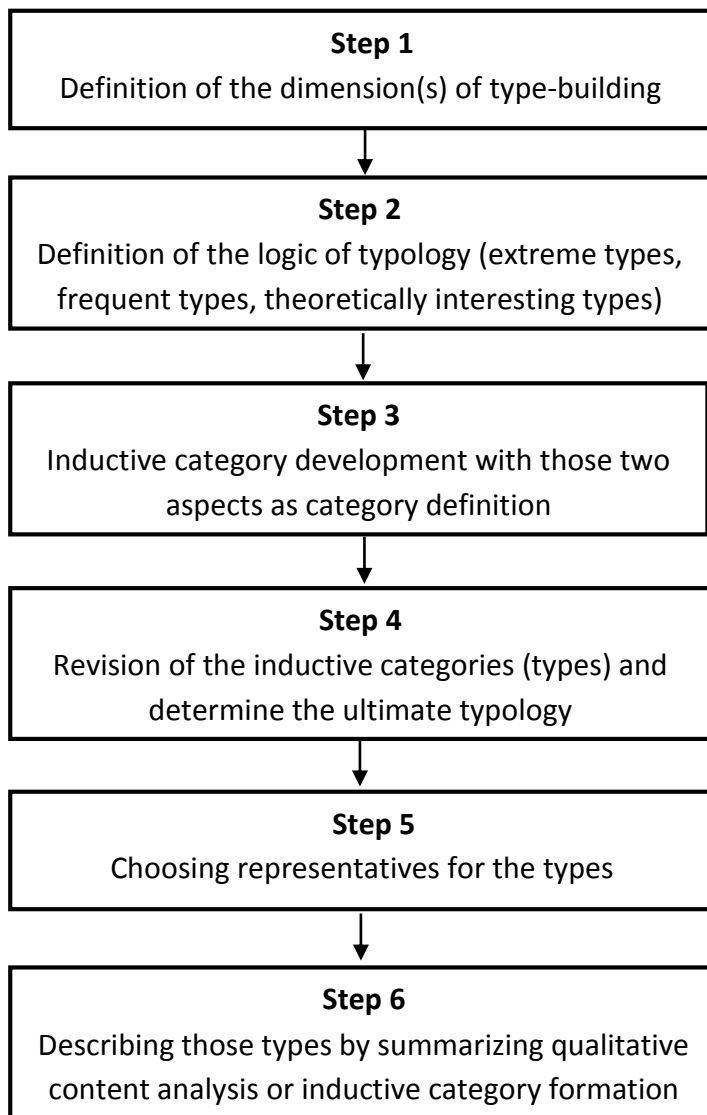


Figure 17: Step-by-step model for type-building content analysis

6.6.3 Parallel procedures

Of course the analysis of the textual material can proceed with different inductive and/or deductive content-analytical procedures simultaneously. In our example study on stress of teacher students we applied the inductive category formation (finding concrete stress factors) and the deductive category assignment (level of self-confidence) parallel in the same passage through the interview material. And several other procedures could be combined in one session. This is the big advantage of content analysis to work through big data amounts very economically.

7. Quality Criteria of Content Analysis

If content analysis is to claim the status of a social scientific method, it must allow quality controls to be applied to it, enabling every individual analysis to be assessed for objectivity, reliability and validity. As far as content analyses hitherto are concerned, however, the position is even more desolate than in the rest of the social-scientific research field: there is an almost complete dearth of data on quality criteria of the procedures.

Koch, Witte & Witte (1974), for example, tested communication science analyses of news media, a classical field of content analysis, with regard to the way in which they treated quality criteria: the most recent six content analyses available to the authors almost all ignore this point. On the other hand it must also be admitted that the classical criteria of reliability and validity are often called into question by content analysts. This point will be dealt with first of all, before quality criteria specific to content analysis are introduced.

7.1 Classical Quality Criteria

Social science methodology divides quality criteria into measures of objectivity (independence of research findings from the person of the researcher), reliability ("stability and precision of the measurement, plus consistency of the measuring conditions", Friedrichs, 1973, p. 102), and measures of validity relating to the question of "whether what is measured is what ought to be measured" (Friedrichs, 1973, p. 100). It is usual to distinguish within reliability and validity different conceptions:

Reliability:

- Re-test: The research operation is carried out a second time and tested as to whether the same findings result.
- Parallel-Test (equivalent Form): The question at issue is examined with the same sample but using a different instrument; then the correspondence is checked.
- Consistency (split-half): The material or the instrument is divided into two equal halves and it is then checked whether both halves yield similar findings.

Validity:

- External criterion: Research findings closely related to one's own issue and objects of examination, and of whose validity one is convinced, are brought in as a standard of comparison.
- Predictability: On the basis of the results predictions are made and then the extent to which they are fulfilled is examined.
- Extreme groups: Parts of the sample expected to yield extreme results are singled out and tested as to whether the results point in the predicted direction.

- Construct validity: The findings are tested for plausibility using established theories and the appropriateness of the operational definitions is considered on the basis of the theoretical background.

Criticism has often been voiced against these "classical" quality criteria and their applicability to content-analytical research (Steinke, 2000; Mayring, 2002a). With reliability determination, parallel testing procedures appear problematic, as the equivalence of two instruments used for analyzing language material is likely to be demonstrable only in rare cases. The splitting method is also unlikely to be appropriate in most instances, since the size of the material sample, as also the size of the instrument (the categories), is mainly defined in such a way that in individual parts central findings can occur which alter the overall results. The usual procedure with content-analytical reliability tests is for the whole analysis to be carried out by several persons and then to compare their results (inter-coder agreement). But objections have been made even to this approach.

J. Ritsert (1972), for instance, points out that a high level of correspondence between different coders could only occur with very simple analyses. "The more detailed and comprehensive the category system is, the more difficult it will be to achieve a high level of reliability in the results, although at the same time the significance of one examination with regard to the contents may rise (transl. PM)." (Ritsert, 1972, p. 70) Lisch & Kriz (1978) doubt the value of inter-coder reliability entirely; believing that with language material interpretational divergences among different analysts will probably be the rule rather than the exception. "Parts of the population that do not view the world and categorize it as content analysts do are simply excluded from further consideration on grounds of stupidity or malice - why, after all, should the social scientist allow his objective significance homogeneity, strenuously achieved with the 'best group of encoders', to be ruined by real reactive and interpretational differences in social sub-groups? (transl. PM)" (Lisch & Kriz, 1978, p. 90).

As reliability is the pre-condition for validity (not, however, the other way round), the arguments against reliability concepts also affect validity. "The stronger the variability of everyday phenomena is determined by undiscovered and/or theoretically disregarded parameters (disturbance factors), the more an increase in reliability through elimination of these parameters will impair the practically relevant aspect of validity (transl. PM)" (Lisch & Kriz, 1978, p. 87).

But criticism of validity concepts is also frequently heard. It is the circularity of validation arguments that is mostly the target of attack (e.g. Ritsert, 1972, p. 72 ff.): when material external to one's own examination is drawn on as a quality standard (external criterion or theoretical assumption in the case of construct validity), then its validity must already have been established. Krippendorff (1980) has formulated this as a trilemma: "If the content analyst has no direct knowledge about what he is interfering, then he actually cannot say anything about the validity of his findings. If he possesses some knowledge about the context of the data and uses it in the development of his analytical constructs, then this knowledge is no longer independent from his procedure and cannot be used to validate the findings. And if he manages to keep the knowledge about the target of his interferences separate from his procedure, then the effort at interfering it from data is in fact

superfluous and adds at best one incident to the generalization of the procedure" (Krippendorff, 1980, p. 156).

It is for this reason that today special quality criteria for qualitative research are under discussion (Flick, 1987, Mayring 2002a, Chapter 5). Such criteria, for instance, are documentation of method, interpretation safeguards, proximity to the object, rule-boundedness, communicative validation and triangulation.

For the solution of such problems, however, special conceptions of content-analytical quality criteria have also been developed. These will now be dealt with in further detail.

7.2 Specific Content-analytical Quality Criteria

With inter-coder reliability a specifically content-analytical quality criterion is addressed. It should be mentioned that the comparison of two analysts coding the same material actually would give a measure of objectivity (independence of research results from the researching persons). Reliability in the proper sense would be the **intra**-coder agreement test, labelled by Krippendorff as stability (see Figure 18). We will come to this later.

Holsti et al. (1969, p. 135 ff.) and also Rust (1981, p. 172 ff.) have pointed out that not only the application of the categories to the material (encoding) must be carried out reliably, but also the construction of the categories themselves. Such considerations are leading increasingly to suggestions for specific content-analytical quality criteria, most recently put forward in their broadest form by Krippendorff (1980). He distinguishes here between 8 concepts, which are connected as follows:

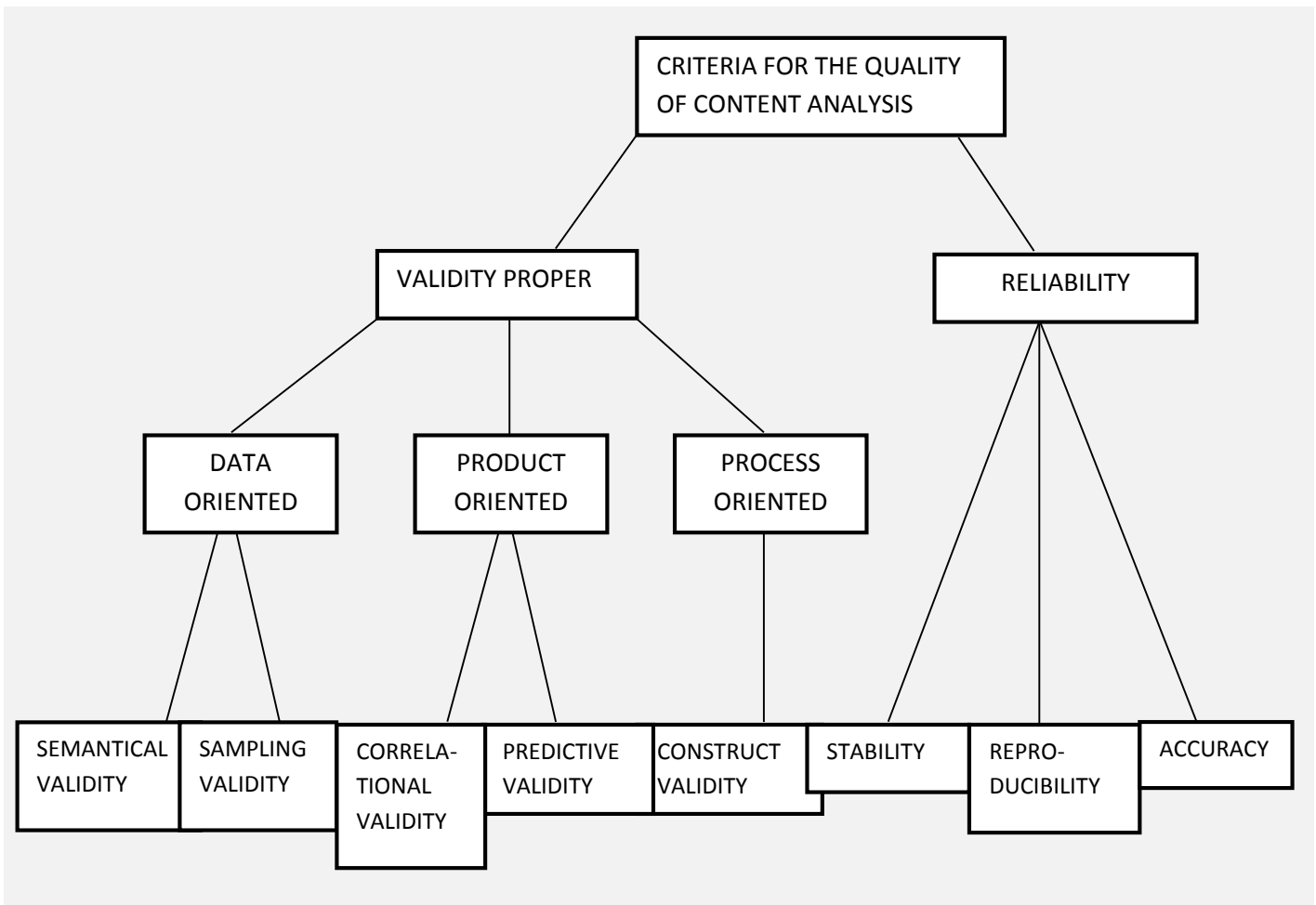


Figure 18: Content-analytical quality criteria according to Krippendorff 1980, p. 158

Semantic validity here has to do with the correctness of the manner in which the meaning of the material is reconstructed. It is expressed in the appropriateness of the category definitions (definitions, anchor samples, encoding rules). Testing can be based on the judgments of experts. But Krippendorff also suggests simple "checks":

- collection of all passages to which analysis instructions have assigned a certain meaning; comparison of the passages with the construct, testing of the homogeneity of the passages
- construction of hypothetical passages with known meaning; testing whether the analysis instrument can reconstruct this meaning; construction of problem cases

For **sampling validity** it is sufficient to refer to the usual criteria for accurate sampling (cf. e.g. Krippendorff, 1980, Ch. 6; see also Ch. 5.2).

Correlational validity means validation through correlation with an external criterion. Testing is only possible if results of an examination with a similar line of inquiry and similar object of study are present. What appear significant are above all comparisons with results arrived at through other

methods such as test, experiment or observation. But the contrary path is also open: often analysis instruments or objects can be named which ought to lead to completely different or even diametrically opposite results. This can also be tested correlationally.

Predictive validity is only applicable as a quality criterion if meaningful predictions can be made on the basis of the material. Testing, however, is then simple and effective.

Construct validity can be tested in content analyses according to several criteria such as

- success rate hitherto with similar constructs and/or situations;
- experiences with the context of the material in question;
- established theories and models;
- representative interpretations and experts.

One quality criterion which is gaining increasing importance should not be left unmentioned here: **communicative validation** (Klüver, 1979; Heinze & Thiemann, 1982), introduced into methodology as well as member check (cf. Flick, 2009). The basic idea of this is to achieve discursive agreement or conformity between researchers and their subjects of investigation (i.e. the interviewees) on the results of the analysis. Such a procedure has a particular "sense and irrevocable necessity, where the theoretical interpretations of statements, especially self-portrayals, have the function of preparing and structuring a research partnership with the interviewees" (Klüver, 1979, p.82). Heinze & Thiemann describe communicative validation as a technique which "(a) contributes to the self-expression of people as regards their everyday lives; it has nothing to do with arguing about the validity of theoretical principles; (b) lends itself to inquiries into the constituent conditions of subjective life; the individuals are regarded at any rate not as simple derivatives of social structures; (c) integrates the most important instrument of research, the researcher himself, into the research process; precisely this is why it is not objective; (d) the research situation integrates co-operation with the daily actors into the interpretation itself; the 'interpretation products' are not separated from the conditions under which they arose; (e) no explanations are given beyond the discussions with the daily participants" (Heinze & Thiemann 1982, p. 641).

Stability can be tested by applying the instrument of analysis again to the material. This is a form of **intra-coder agreement** and a measure for reliability in the traditional sense (comparable to retest-reliability in test theory). It is very easy to accomplish and therefore highly recommended within qualitative content analysis: After the coding process the analyst starts again with coding from the beginning of the material without knowing his or her previous codings, at least for a part of the material. Then he or she compares the two results. This gives insights if the rule application had been stable during text analysis. If the results are very different, the rules (units, category definitions, abstraction levels, coding agendas) should be revised and all the material should be analyzed again. If there are only small differences, this should be reported as measure of reliability.

Reproducibility means the extent to which the analysis leads under different circumstances to the same results. This factor depends on the explicitness and accuracy of the process description, and can be measured via **inter-coder agreement**. Usually this procedure is labelled as inter-coder-reliability, but we would say that it is more objectivity in the sense of independence of the results

from the analyzing person. The simplest measure would be the percentage of agreement (identical codings divided by all codings). But there are a lot of more specific suggestions of coefficients (for a survey, see Friede, 1981; Asendorpf & Wallbott, 1979). Such coefficients must not only account for the proportion of correlating assessments by different coders, as in the measure of reliability (Holsti, 1969, p.140):

$$R = \frac{(\text{Number of coders}) \times (\text{Number of correlating assessments, agreements})}{(\text{Number of all encoder assessments})}$$

They also ought to rid the coefficients of the number of expected chance correlations, as Scott, Flanders, Garrett and Cohen (cf. Friede, 1981) attempted.

$$R = \frac{(\text{observed percentage agreement}) \times (\text{expected agreement by chance})}{1 - (\text{expected agreement by chance})}$$

Krippendorff (1980, p.133 ff.) produced a coefficient which seems to be the most suitable. He starts from the following basic idea:

$$R = 1 - \frac{(\text{observed coder disagreement})}{(\text{expected disagreement by chance})}$$

Krippendorff has worked out this approach to inter-coder- reliability for several encoders, several features and all scale levels (nominal, ordinal, interval, and ratio scales).

Accuracy refers to the extent to which the analysis conforms to a particular functional standard. It presupposes the stability and reproducibility of the instrument, it is the strongest measure of reliability, but at the same time is the most difficult to test.

According to Krippendorff four sources of non-reliability can be distinguished:

- the assessment units (discovery points): here it can be tested whether the assessment units where discrepancies between several encoders occur are systematically distinct from the rest of the material;
- the analyst: this can be tested via inter-coder reliability;

- the individual categories: here it can be tested whether discrepancies occur with particular frequency in the case of certain categories; this can be eliminated by making the definitions clearer;
- category differentiation: reliability can often be increased if ambiguous categories are amalgamated, thus leading to a category system which is more general, but more accurate in its applicability.

This conception of Krippendorff constitutes a version of content-analytical quality criteria which is rational and, for the most part, easily applicable. Systematic compilation of quality criteria ought, however, to start with a content-analytical theory of error. The question that should be asked is: Where can content analyses still make mistakes? Quality criteria would then be related to this. Material on such a theory of error could be found in two areas:

- In the object model, the content-analytical communication model (cf. Figure 8) the relation between the material, its subject matter, the communicator, the recipient and the content-analyst is portrayed. Distortions can arise between all these entities. They can be further categorized as sources of error.
- In the procedural model of analysis (cf. Figure 9 in general) the individual analysis steps are described in sequence. Every one of these steps describes at the same time a possible source of error.

Reflection on possible content-analytical sources of error could lead not only to the development of new quality criteria; the suitability of content analysis as a social scientific method in general ought to be established here. For us, a check of intra-coder and inter-coder agreement (at least for parts of the material would be the most important concepts, and actually indispensable for Qualitative Content Analysis.

Link to QCMap software (www.qcamap.org):

In QCMap on the screen of the project details a button “+ New Inter-Coder-Agreement” is visible. With this tool a second coding is made possible. It is recommended to run this comparison of the two coding processes with the same material as intra-coder comparison (stability) and inter-coder comparison with a second person (objectivity).

7.3 Three Levels of Inter-coder Agreement

For Quantitative Content Analysis the calculation of coefficients like Cohen's Kappa or Krippendorff's Alpha play an important role. Coefficients should be higher than 0,8 with a minimum for acceptance of 0,67 (Krippendorff, 2004). In qualitative research however, a perfect agreement between different analysts can hardly be reached, because interpretative elements (even if extreme rule guided like in Qualitative Content Analysis) always bear a subjective element. So we must be a little bit more modest. To leave out inter-coder comparisons would not be an alternative, because it leaves text analysis in pure subjectivity.

We suggest three different levels of inter-coder agreement tests which are different in their degree of rigor:

- The strongest test would be to give only the texts to be analyzed and the research question(s) to a second person. So we can check if the process of category building, category definition and category application, as well as the definition of procedures and units of analysis is the same. But within those definitions a lot of theoretical considerations are introduced, and research results always have to be seen as theory dependent.
- So a second way would be to give to a second coder the texts to be analyzed together with all content-analytical rules (procedure, units, category definition and level of abstraction for inductive category formation, coding agenda for deductive category assignment). This is the best way for most of Qualitative Content Analysis projects. But sometimes the material is very open, no theory can lead to clear definitions, and the research question is widely explorative.
- In those cases a "lighter" test is recommended. The second coder has insight in the whole material, definitions, and codings of the first coder. He works as supervisor and checks if he or she can confirm the analyses of the first coder.

The project should decide for one of those procedures. And of course several coders (more than two) can be involved.

Because those procedures can be very time consuming sometimes only a partial inter-coder agreement test is carried out. Only parts of the textual material (random samples, exemplary text portion, difficult text portions) are selected.

A further specificity of Qualitative Content Analysis is the possibility of correcting false codings, especially if the text corpus is not so huge and the inter-coder agreement test is run through the whole material. This is a unique possibility to come to better results, instead of only having an indicator for accuracy. All codings with disagreement could be excluded from the further analysis. Even better would be to organize a sort of coder conference, where the coders discuss the disagreements and decide for the right codes.

Link to QCMap software (www.qcmap.org):

In QCMap the three possibilities are offered on the screen and the analyst has to decide for one of them. After running the inter-coder agreement test, not a quantitative indicator is offered but an open appraisal of reliability or objectivity.

If a quantitative indicator (Cohen's Kappa or Krippendorff's Alpha for example) is needed, the results must be exported via the analysis screen and imported into a statistical software package.

8. Computer Programs for Qualitative Content Analysis

There are two reasons for thinking about the use of computer programs for Qualitative Content Analysis: First, the textual material nowadays usually consists of a text file which makes it possible to transfer it into a software program. Secondly Qualitative Content Analysis represents a very systematic, controlled, step-by-step sort of text analysis, where maybe a computer program could be helpful.

And indeed since the eighties of the twentieth century a lot of programs have been developed, especially for qualitative text analysis, nowadays under the label of CAQDAS (Computer Assisted Qualitative Data Analysis; Pfaffenberger, 1988; Tesch, 1990; Fielding and Lee, 1991; Weitzman & Miles, 1995).

In the context of qualitative research computer programs play a different role as they do in quantitative analysis (see chapter 3.1). Looking at recent developments, the following computer procedures (Kuckartz, 2005) are relevant for qualitative content analysis:

- The textual material is transcribed using a word processor, so that we can read the material as a text file within different computer programs (e.g., as ASCII file). The specific program can edit and organize the material for the different procedures of analysis.
- We can mark specific segments of the material ("underlining") and attach keywords or categories to them (coding). Some computer programs do this by referring to the line numbers, some by using the mouse, others by using hypertext functions.
- We can mark other segments of the material and subsume them under formerly defined keywords or categories.
- Now we can gather all material coded with a specific category, even from huge quantities of text. This allows us to single out typical quotations for individual categories.
- We can pick out segments of text marked with keywords or former coded categories.
- The categories can be altered, revised and refined in the process of analysis.
- The categories can be ordered hierarchically, divided into subcategories, combined into general categories, together with all associated text segments.
- Rules of analysis, comments on the material, and explanations of categories can be attached to the categories within the computer program, so they are available and revisable at any moment within the process of analysis.

So in qualitative research the computer has totally different functions from those in quantitative research:

- The computer serves as an **assistant** to the researcher. The researcher is still responsible for the interpretation of the text, but the computer helps to organize the materials, the steps of analysis, the interpretation rules, and the results.
- The computer is the **documentation center** of the analysis. Every decision of organizing, coding, and interpretation of the material is "written down" and can be reviewed at any time

in the process of analysis. The fact that everything is documented also gives one the opportunity to reconstruct, at a later date, the situation in which the interpretations were formulated. This is important for reliability checks.

- Under certain conditions (e.g. within structuring content analysis) the computer can **prepare the results** of analysis for further **quantitative** processing. Some computer programs for qualitative analysis provide simple quantitative procedures themselves. Otherwise the results can be copied as a data file into a quantitative program and thus can be combined with other quantitative data. This is of course valid only if the qualitative analyses produce results which can meaningfully be quantified (e.g., frequencies of the occurrence of categories).

In recent years several computer programs have been developed which can be really helpful for qualitative content analysis. Weitzman & Miles (1995) discuss 24 different programs for steps of qualitative analysis which can be adapted for qualitative content analysis (e.g. ATLAS/ti, MAXQDA). In the meantime the label CAQDAS (Computer Assisted Qualitative Data Software) has been introduced and several internet pages collect and comment the latest software developments (e.g. <http://www.surrey.ac.uk/sociology/research/researchcentres/caqdas/>).

There are some limitations of those programs: Most of them are developed from the background of Grounded Theory. They offer the possibility of coding, code networks, and memos. Other approaches of qualitative text analysis are not so easy to apply. For deductive category assignment for example it would be important to have the coding agenda on screen during coding, for inductive category development the category definition and the level of abstraction. Within the traditional programs the memo function can be used for that, but there is usually no possibility to create tables (for the coding agenda). A second limitation is that the traditional programs are constructed in a window design. The screen is divided in different windows (e.g. a window for the texts, a window for the codes, a window for the memos). This is a more or less static concept, and the screen can be overcrowded. Especially for Qualitative Content Analysis we need a definition of units of analysis, step models, category definitions, coding rules, and so on, which could hardly be displayed on the screen (different memos for that?). So Qualitative Content Analysis can be proceeded, but not very comfortable.

Within the last years, funded by the Alps-Adria-University Klagenfurt, the Kaerntner Sparkassenfonds, and the Association for the Support of Qualitative Research ASQ, a software package for Qualitative Content Analysis (QCAmapping) has been developed. The software engineers (coUnity Software Development, Klagenfurt/Austria) had suggested, and I think this is the first time in Computer Assisted Qualitative Data Software, an interactive web application, which opens step by step new screens, following the methodology of Qualitative Content Analysis. If one of those steps is not proceeded (for example no units of analysis or coding agenda defined, no pilot study) the program stops.

This application is offered for free via open access at www.qcamap.org.

Because program refinements are done continuously an online solution has been selected instead of a download solution, installing the program on the individual computer. So we can improve the

program, add new possibilities, correct failures without the necessity for the users to download new versions. The program is kept on an independent, highly protected server. An additional homepage gives actual information (e.g. workshops, projects, publications) around the program (www.qualitative-content-analysis.aau.at).

The following slides give an impression of the program.

Starting your work

- First you have to create a new project (title and description)
- Then you have to define a research question and description!
- Then you chose the content analytical technique (currently, inductive category formation and deductive category assignment are available)
- You should give a description of the research question!
- By pressing „Create“ the program will save your research question and leads you to the next step, where you have to specify details for the chosen content analytical technique.

The screenshot shows the 'Add Research Question' form in the QCAmap application. The form has a title bar 'QCAmap' and a navigation bar with 'Home / My Projects / Test / Add Research Question'. The main heading is 'Add Research Question'. There are three input fields: 'Research question' with the text 'Description of stress factors in first praxis experiences', 'Content analytical technique' with a dropdown menu showing 'Inductive Category Formation' and 'Deductive Category Assignment', and 'Description' with the text 'First professional experiences, especially for teachers, are often described as "praxis shock" (Smagonsky et al. 2011; Mueller-Forbrodt, 1978). We want to describe the concrete situation further'. At the bottom are 'Create' and 'Cancel' buttons. Arrows from the text on the left point to each of these three input fields.

Upload your text material

- For each research question you have to upload the text documents which will be analyzed!
- The text documents have to be transformed into a .txt.-file (UNICODE-Format; the procedure how to convert your files is described on the entry page under the button „How to get started“.

The screenshot shows the 'Project details' page for 'Example Teacher Unemployment'. It includes a 'Research Questions' section with a description of stress factors in first praxis experiences as teacher. Below this is a 'Documents' section with a table listing four files: Case A.txt, Case B.txt, Case C.txt, and Case D.txt. A blue arrow points from the 'Upload new Documents' button to the 'Documents' section. Another blue arrow points from the 'How to get started' button in the 'Introduction' section to the 'Start Coding' button in the 'Documents' section.

#	Filename	
1	Case A.txt	
2	Case B.txt	
3	Case C.txt	
4	Case D.txt	

Introduction

- With a series of slides we would like to show you the basic procedures of the software.
- The program entry page (www.qcemap.org)
- The Website with further infos (www.qualitative-content-analysis.eu.at)
- Informations about first steps to get started
- You need to register an account. We guarantee that all your data will be kept confidential. A registration notification will be sent to your email-address.
- Now you can use the software from everywhere using different webbrowsers (Internet Explorer 9, Mozilla Firefox, Google chrome).
- Do not forget to log out. Your data is stored on our server. We guarantee that only you have access to this data.
- Since the application is web-based you will always work with the latest version of the program.

The screenshot shows the QCAmap website homepage. It features a navigation bar with 'Home', 'My Projects', and 'Example Teacher Unemployment'. The main content area includes a 'Project details' section, a 'Research Questions' section, and a 'Documents' section. A blue arrow points from the 'How to get started' button in the 'Introduction' section to the 'Start Coding' button in the 'Documents' section. Another blue arrow points from the 'How to get started' button to the 'How to get started' button in the 'Introduction' section. A third blue arrow points from the 'How to get started' button to the 'How to get started' button in the 'Introduction' section.

QCAmap is software for Qualitative Content Analysis. It is an open access web application for systematic analysis in scientific projects based on the techniques of qualitative content analysis.

QCAmap can be used with research projects in e.g. Psychology, Sociology, Education, Economics, Literature, etc. to analyze small and large amounts of any text material coming from interviews, group discussions, observation protocols, documents, open-ended questionnaires, etc. **Qualitative Content Analysis is a strictly rule-guided procedure containing qualitative steps** (assignment of categories to text passages) **and quantitative steps** (analysis of category frequencies).

Literature

Mayring H. (2010). Qualitative Content Analysis. In: Mayring H. (ed.), Qualitative Content Analysis. A Guide to Its Methods. Berlin: Springer.

Mayring H. (2010). Qualitative Content Analysis. In: Mayring H. (ed.), Qualitative Content Analysis. A Guide to Its Methods. Berlin: Springer.

Deductive category assignment

- If you chose the technique of deductive category for your research question, you have (after defining the content analytical rules) to formulate the coding guideline, which contains definitions, anchor examples and coding rules.
- After coding a certain amount of material the program reminds you to revise the category system and the coding guideline (pilot phase). After finishing the pilot phase, no more changes can be applied to the coding guideline.

UCAmap Filtered

Edit Research Question

Research question: Has "practice shock" affected the self-confidence of the interviewee?

Content analytical strategies: Deductive Category Assignment [Step model & rules](#)

Description: The question is: what makes the first practice experience for the person himself. Does he or she grow or loses self-confidence? The interview material on practice shock has to be added to research strategy document self-confidence.doc

Content analytical units

Research question is already revised.
Modification of content analysis of units and the coding guideline is needed.

Coding unit: Clear semantic elements in the text

Context unit: The whole interview, interview protocol and background material

Recording unit:
 ☐ 1 document
 ☐ After merge categorization
 ☐ Count multiple codings

Coding Guideline

Name	Definition	Anchor Examples	Coding Rules
C1: High self confidence	High subjective feeling of solving the challenge	Of course there were also problems, now and then, but they were simply solved, owing to a change	All three aspects of the definition must point in the direction of "high", at least no

Inductive category formation

- You can look up the step model and the content analytical rules of inductive category formation (Mayring, 2011; 2013)
- If you have decided for inductive category formation, you have to describe the research aim, that is you have to define the content analytical units (recording unit in inductive CA is always the entire material), to give a category definition and to determine the level of abstraction!
- You can revise those definitions during the pilot phase (do not forget to save the changes with the button at the end of this page!).
- Only once these definitions are given, the program will allow you to „Start Coding“ your material!

QCAmap

Research question Description of stress factors in first praxis experiences

Content analytical technique Inductive Category Formation
[Step model & rules](#)

Description First professional experiences, especially for teachers, are often described as "praxis shock" (Smagorinsky et al. 2011; Mueller-Forbrodt, 1978). We want to describe the concrete stressor factors.

Content analytical units

Coding unit Clear semantic elements in the text

Context unit The whole interview, interviewer protocol and background material

Recording unit All documents
☒ Count multiple codings

Category definition

Category definition Stressful experiences in and around teaching, experiences of harm, loss or challenge which are not automatically coped (Lazarus)

Level of abstraction

Level of abstraction Concrete stress factors for the person, connected with negative

Coding

- You can start coding. The program always shows you the research question and rules on the lefthand side, which also can be revised during the pilot phase of coding.
- Mark text passages with the cursor and formulate a new category or subsume the passage to a previously formulated and thus already existing category!
- The list of inductive categories will be growing.
- During the pilot phase you can also edit the category system.

Coding

- During coding the texts you can always have a look on the coding guideline.
- Mark text passages with your cursor and assign them to one of the categories in your coding guideline!

- You can run several research questions (inductive and deductive) within the same project, with the same text material.
- When your analysis of the texts is advanced, the software remembers you to pilot check the category systems (inductive) or the coding guideline (deductive) and the content analytical rules.
- Once coding of the text material is finished (but also during coding) the system gives you several possibilities of displaying the results (button „Analysis“).
- The results will be given as Excel-files, so the data can be imported into statistical analysis software (e.g. SPSS).

The screenshot shows the QCAmap web application interface. At the top, there's a navigation bar with 'Home', 'My Projects', and 'Example Teacher Unemployment'. Below this is a 'Project details' section for 'Example Teacher Unemployment', which includes a description of the project and a 'Use Research Question' button. The 'Research Questions' section lists two questions: 'Description of stress factors in first praxis experiences' and 'Has "practice shock" affected the self-confidence of the individual?'. Each question has a 'Start Coding' button, an 'Edit' button, and a 'New Item/Code/Agreement' button. The interface is clean and professional, with a light blue and white color scheme.

- Enjoy the program!
- New features will follow soon.
- If you have questions or problems, or if you find any features that do not work properly, do not hesitate to contact us. Your feedback is very much appreciated!

This screenshot is similar to the one above, showing the QCAmap interface. It highlights the 'Documents' section at the bottom, which includes a 'Upload new Documents' button. The 'Research Questions' section is also visible, showing the same two questions as before. The interface is consistent with the previous screenshot, maintaining the same layout and design elements.

9. Related text analysis approaches

First I want to compare the procedures of Qualitative Content Analysis with similar approaches of the qualitative oriented social science text analysis (cf. Mayring, 2010b).

Within media analysis, David Altheide (1996) has developed a procedure ("ethnographic content analysis") working with deductive categories (codes), which were refined in the process of analysis. Then he summarizes the results for each category. This has similarities with our approach but is not at all as rule oriented as Qualitative Content Analysis. In the USA there exists an approach coming from the quantitative content analysis which is called Codebook Analysis (Neuendorf, 2002). It is a deductive category application procedure, which defines in the codebook all categories and gives examples from the text. But this definition is not as systematic as the coding scheme (definitions, anchor examples and coding rules) in our procedure. In some ways similar is the Thematic Text Analysis (Stone, 1997), which looks through the text for central themes, using theoretical preconceptions or empirical word frequencies and word contingencies. In both cases the Qualitative Content Analysis defines the procedure more precisely. The related concept of Theme Analysis covers phenomenological procedures more freely (Meier, Boivin & Meier, 2008). Some similarities can be found between Qualitative Content Analysis and text analysis following Berg (2004). He describes deductive ("analytic") and inductive ("grounded") categories which have to be defined explicitly, but it remains unclear how this has to be done. Schreier (2012) describes techniques of qualitative content analysis widely based on our developments. She first introduces a data-driven coding frame (we would call it inductive category development), but there are some misunderstandings of our concept in regard to building new categories and subsumption of material to just formulated categories. Then she describes a concept driven way of coding, similar to our deductive category application. Kuckartz (2014), the developer of the widely used software program MAXQDA, describes three different procedures of Qualitative Content Analysis, again broadly based on our developments: thematic qualitative text analysis (cf. above), evaluative qualitative text analysis (in analogy to our deductive category assignment), and type-building text analysis (see above). We think that this concept is selective, taking up only some possibilities of Qualitative Content Analysis (for a broader discussion see Mayring, 2014).

In comparison to those text analytical approaches the Qualitative Content Analysis seems to be the broadest (describing a wide set of different procedures) and most exact one (prescribing clear step-by-step models and analytical rules). So Steigleder (2008) after a praxis test of qualitative content analysis comes to the conclusion, that "it has proven its worth in many studies. With its different techniques of analysis and its methodological concept it is excellently adapted to analyze qualitatively collected material" (Steigleder, 2008, p. 197). But it should not be argued that Qualitative Content Analysis is the only legitimate text analysis procedure. It depends on the concrete research question and the quality of the material, which procedure should be chosen. If a use of the strict category relatedness and rule orientation of the Qualitative Content Analysis would

neglect important deeper aspects of the material (e.g. repressions in the sense of psychoanalysis), then other procedures (e.g. psychoanalytical text interpretation) would be more adequate.

Appendix

Excerpts from semi-structured interviews with four unemployed teachers, carried out within the framework of the research project "Teacher Unemployment" (Ulich et al., 1985).

Q = Questioner

I = Interviewee (i.e. the teacher)

Case A:

I: Well, it certainly **wasn't a strain for me, at least from the, well, the physical side of things. The contrary in fact. I was sort of pretty keen to get down to teaching at last.** You're studying, you see, for the teaching certificate and that is your course, the academic part of it, I mean, up to the First State Examination, that is... that has nothing to do with teaching as such, and in my practical - we have to do a sort of practical - and I had the luck to be able to teach a full two weeks- that was the time I was there - at a senior elementary school. Normally all you do during these practicals is sit in on other people's classes - just sit at the back; which is incredibly boring of course just listening to someone else teach for two whole weeks. And it so happened that at that time they were a bit short of teachers and the principal says to me: "Listen, I know what we'll do. You take the 8th and 9th grades in physics and mathematics, then I don't need to do that myself any more; that's extra work for me, you see and if you do it, I'll have more time for my administrative stuff."

Q: So that was still during your undergraduate period?

I: That's that's the same for everyone. The practical has to be done by everyone at a high school, senior elementary or junior elementary school. So I was able to teach two whole weeks there and I had a marvellous time. Senior elementary school is of course relatively simple as far as preparation is concerned, as the content is not so difficult. In 9th grade maths there's Pythagoras, well...

Q: We know!

I: ...which you can do more or less straight off if you're a science student, and to the students you're a magician anyway when you give them a demonstration with the circle of Thales. They say, "That's incredible, it's almost magic!" And that's what I enjoyed. that's why **I was already looking forward to being able to teach at a seminary school. Certainly, there are disappointments that the students are not as one thinks they ought to be. I mean, in a big city like this there are just a lot of problems, what with the big firm here. And it is certainly not as you really thought it was going to be, but well, it was certainly not a practice shock for me.**

Q: Hm. Not even with the large number of teaching hours you have, with preparation - and, well a lot have told us this - that the postgraduate training phase is terribly stressful, with all the work.

I: Well, I... on that point you have to remember that in the first period, the first half-year in the training school and in the third period as well, you have to remember that as a rule you only have one class per subject. So as a rule that means between 4 and 8 hours. O.K., someone with English or French might have ten hours - 5 hours English, 5 hours French in one class. But that's only 2 classes as a rule. So that's not such a problem. Where it does get a bit problematic is in the branch school, and that's probably what the people meant, there, in Bavaria postgraduate trainees work for hours...

Q: Yes, exactly.

I: ...per week, one day off. The training regulations stipulate one free day per week, if possible, which should be Monday or Tuesday. Let's say you have one day off, then there are four days left; so with sixteen teaching hours a week you average four hours a day. You certainly have work to do, no doubt about it. The point is, though, it tends to vary from school to school. If you're at a school, for instance an inner city school where you've got discipline problems, where the students just - are completely different personality-wise, then maybe you do get somehow frustrated as a teacher. But in my case at the country school - and I had the maximum number of hours, eighteen per week, and had a great many classes, a very great deal of preparation in other words, and completely - different, varied, whether it's a sixth grade class in geography or an eleventh grade in geography, that makes a difference you know. So I got to know practically all the students at the school, from the children to the adolescents. But for me that was a great, sort of, compensation - **I didn't even notice that I had so much work because I enjoyed it so much. because I saw that the students enjoyed it too.** And a little while ago I went back to the school, and you should have seen them as I arrived at the school building, immediately came running up to me when they saw me and said "Well, how are you, and are you coming back to us now?" and things like that. And they, well for me, for my part, I was really over the moon, and they said, "Oh dear, we've got such and such a teacher now and it's just not the same as it was with you," that sort of thing. And I said "Yes", and "Where are you now?" they asked. And when I said, "I'm unemployed" - "Incredible, we don't believe it, how could you be unemployed?"

Q: But...are they, weren't there any problems caused by the fact that when you're teaching you can't properly adjust to the children. You have rather a lot of students and - well, a lot of teachers have told us that you imagine beforehand a lot about how

you're going to devote time to each student and that people are then disappointed by what school life is really like.

I: Well, it may be that I didn't really have any advanced ideals or anything, because I'm just too realistic, I mean, if I've got a class of 35 students and I have one lesson of 45 minutes with them each week then I can only begin to think about applying general educational principles in a very very small way. I must, sort of, or rather I certainly use the, well not just the approach, but I aim for a high level, of course, in my whole work, but of course what comes out at the end is very little, because 45 minutes, 35 students, that means for each student I've got a time "ration" of about one minute. And I mean, I can't even use that time to deal with real educational issues because I have to get through the syllabus.

Case B:

I: Practice shock, yes, I used it in the thesis that I wrote too (laughs), I must say I've never really suffered from it - not directly. I was roughly aware - **you start the job really with a very positive attitude**, don't you, something like sort of "Here I am, this is me!". I know I told a friend of mine who's just finishing what I was doing in class, in the sports lesson. And he said, "No, impossible, there must be another way of doing that." And I said, "Ah, that's what I said when I was still studying," I said, "You're all incompetent, just wait till I get there and you'll see", and that with the methods you mean, - above all with talking to them, encouraging them, saying things like "That was stupid, don't do things like that" it is really only possible in the rarest of cases." And I must say, **with the class I had initially, eighth graders, a bit of a difficult lot, I managed in the course of the year to get onto a good footing with them** and I never really suffered from any shock, I must say. **I just took things as they came** - and I particularly noticed very quickly that the others, the experienced teachers, had the same difficulties I had, that was it, you see. And if they have difficulties, I thought, well, then I really don't (laughs) need to get uptight about "Am I a failure or am I not a failure?" Do I, really? There are very few teachers who admit to having problems, I mean there are also very - at the schools I've been at so far - there are also teachers who're nothing but successful, apparently, and then you happen to be passing the classroom one time, and you hear the rumpus, the racket going on, and then you know. But - I was at a high school, and the staff there were very young, open for new ideas, and even the older ones, those who'd been at school for 6, 7 years, would come up to me and ask, "How would you do this?. This or that student behaves so badly, how were you taught in your training to deal with things like that?" And I felt that was great, really great, that people were willing to converse openly with one another and to say "Listen, I've got problems with this or that student, what do you do and how does he behave with you?" And that's what I found was the best solution for practical problems that occurred. But I had no direct practice shock. I'm really very flexible (laughs), and if anything shocking really happened, then - or whenever it did happen - then **I always knew how I should react**. Whether it was educationally valuable, my reaction, whether I could have reacted differently, well, things always look different with hindsight, don't they; but at that particular moment, especially in a sports class, shouting (laughs) is often much more useful than going up to the person and talking to them intently, because then it's probably too late...

And then particularly in class, I always had very large classes, you see, in geography above all, 30 was the smallest number, but I often had around 38, and that's, I mean, then there really are situations arising which cause problems, I mean, you're talking to one student and the rest just start messing about, you talk to

another student, and...that's...you have hardly any chance, you see, and you're really forced then to do things, act in a way that - (laughs) to be really honest - I could never have imagined behaving, and the stupid thing is, you're in that situation, the situation is there, has arisen, the students expect you to do something. You've got to react and - things get done and said of course which I think to myself now, "Jesus, that was ridiculous the way you dealt with that one", or "Wouldn't it have been better to have reacted in such and such a way?" But...then...

Q: Is the way you learn to deal with discipline problems in the postgraduate training phase adequate? Or is that...

I: Well, ...I...must...you learn, you adopt or develop a certain, well, repertoire, I suppose, of certain reactions to situations. Quite honestly, I must add that a principal once pointed something out to me, he said, "Take the postgraduate phase as an experimental one. Try out whatever there is to try out. When you're a fully fledged teacher later, you'll find you have an image you're tied to. You can't just say 'Well I'll react like this today and like that tomorrow!'" So I always had that in the back of my mind and tried out a few things. Somebody once said: "Just give the desk a decent thump with the atlas, then you'll get some quiet!" And then a situation arose where I thought, "O.K., that's what you're going to try right now!" And, well, it worked. Not for very long, mind you, but I just always tried out these things, these tips that I got from other people or had thought out for myself in answer to the question "What do you do when this or that situation arises?" And to that extent I've always worked at self-improvement and said to myself: "That was acceptable" or "That was a bad move, that one." Of course, you've got to be consistent and push on to the bitter end. You can't say "That has failed, time for a retreat." That is - it depends on the classes, but the classes I was in, I believe, wouldn't have tolerated retreat. That would just have meant emptiness, nothing. I mean you're certainly in a dilemma there a bit, aren't you. But a lot can be learned from a student's behaviour. And I must say, I've always basically had - only the negative points are coming out now - good relations with the students. And that partially comes from the fact that I was often with them on school ski-trips and then you get to know each other in a completely different way - and in sports classes you have a different relation to the class anyway, one that you can't really compare, I don't think, with the atmosphere of normal classroom work. In geography it takes a bit longer. Firstly you sometimes have only one hour a week in the subject, and if you don't also teach sport in the same class, then (laughs) then after half a year you know..., you can think yourself lucky if you know the students by sight and can sort of at least place their faces (laughs). But of course that's not very much, and then the contact to the class isn't so good. -

Case C:

I: Yes, that is a big problem, no doubt about it, but of course you must bear in mind that with, during the postgrad training phase everyday school life only partly falls on one's own shoulders, somehow - so you're always aware...because primarily, I'd say, you see the function, or I should say your own dependence on the seminary instructors, that's the thing that is the really dominant factor first and foremost, - during the university course there was this education practical, the practical credit you have to do, so that you have to go to school and do a few..., that's been made considerably more intensive now, I think.

Q: Hm, yes.

I: They didn't have that in the past and you only went into one or other class for two weeks or so, mostly only as an observer, and then you held a lesson, and I thought, Jesus, that's pretty meagre, really, that's not what I expected, it really ought to be done like this and that, but on the point that - this idea that you sort of develop, you can't put that into practice at all during postgrad training, I've seen it myself, - I didn't really see it myself that way, of course, more - the role I was playing more, like all the others too, I suppose, - all the other trainees too, that we were all concentrating really - somehow **looking to get assessments which were as good as possible**, and everybody tries there somehow to (laughs) work out a plan, or thinks he has worked one out.

Q: Hmm.

I: A plan or an idea of how he can best fulfil the expectations of the seminary instructor and that of course **leads to a conflict situation if you, - yourself really wanted to be doing something else in that situation, but because of these external criteria, which are quite openly assessed.**

Q: Hmm.

I: By the seminary instructor. - teacher, this - this, well, this or that action is not appropriate and therefore things shouldn't be done like that, as much as if to say, putting it in clear terms, that you have to fit in from the beginning with what the seminary instructor has in the way of ideas and policies. And that was not so...

Q: And did you have problems on that point?

I: Pardon?

Q: Did that create any problems for you, was it difficult for you?

I: Oh yes, it was, because I didn't, because I'm not really the type that can apply mechanical rules from the beginning, right from the first meeting with a new class.

Q: Hmm.

I: Instead of that and despite everything, one looks to develop some sort of relationship to the students, which means that as a consequence your reactions, or rather one reacts in some cases perhaps differently from the way one is somehow expected to react according to the official credo. One reacts partly in the way one thinks fit, although of course occasionally one makes mistakes (laughs).

Q: Hmm.

I: It might also be the case, that this, and I quite admit this, that this might not be - well, speaking objectively, perhaps not quite so pronounced as I thought myself. **Perhaps I'm a bit, well perhaps a bit over-sensitive on that point,** I suppose.

Q: Hmm.

I: But I do know from conversations with other trainees at the seminary that most of the others felt the same way as I did.

Q: And that has particularly to do with the ideas the seminary instructors have?

I: Yes, with their ideas on the one hand and then - with the permanent feeling or awareness you have that you must get a good mark, as good a mark as possible.

Q: Hmm.

I: And - you get told in more or less as many words that the average mark must somehow be adjusted on the basis of how you shape up generally with your instructors. So the marks are not just a reflection of what happens in the classroom itself, but of how you're assessed verbally, so that of course you - you try for all you're worth to get as good an assessment as possible.

Q: So that is what you might call "pressure to conform"?

I: Yes, certainly, that's the way I see it, yes.

Q: Hmm.

I: It could of course be that in the immediate future that will lose some of its force, as everybody knows there will be no chance of employment (loud laughter).

Q: (Also laughs) And how was it solved, that problem?

I: **I'd say, right up to the end,** the oral in the Second State Examination, which only counts one-seventh of the total postgrad phase, I'd say that the problem was with us right to the bitter end, really.

Q: Hmm.

I: Well, all right, after the third lesson test it had probably disappeared.

Q: Hmm.

I: Up to the autumn in other words.

Q: Did it in any way take its toll on your nerves?

I: Yes, it did, I must say. The opportunity is also given you to repeat the postgrad training phase if you want a better mark.

Q: Yes, yes.

I: Then it would be counted again, - but quite honestly I'd never have been able to do that, just from a psychological point of view.

Q: Hmm.

I: Especially as there's be no advantage in terms of formal calculation, as it only counts for two-fifths of the total mark.

Q: That of course isn't very much.

I: And I mean, I've got no illusions, grade one lesson tests are not within my province and the impression that I'd never get a one, that makes the whole thing illusory. In any case you can only improve your total mark by a maximum of three-tenths, I think. So you're not helped much by improvement in just one single assessment mark.

Q: Hmm. Did that in any way affect your self-confidence, this pressure to conform?

I: - **(in a small voice) Yes, I suppose it did really,** - it's sort of, well, it lost some of its edge in everyday school life, I'm not too sure how to put that (laughs) - I didn't, well, it wasn't like what you had thought in advance, I mean the ideas you had, it wasn't as if they were completely wrong, or you said to yourself "I can't deal with children after all", I mean I didn't come to that conclusion, - it's more, I think - **it eats away at you, and for that reason - makes inroads into your self-esteem,** I suppose.

Q: Hmm.

I: Although it varies according to what type you are, I think. Some are not so bothered, they put on more of a face, they regard it more as, let's say you could see it this way, that the educational qualities they already have, though I'd put "educational qualities" in inverted commas, that they say to themselves, well, it has to be done like that, it has to be done like that, and then they do it like that. And if they're lucky it goes well for them, precisely because they've done it like that, and that's all right, isn't it.

Q: Hmm.

I: This might be a bit of an exaggeration, but I'd say it's very important, especially in sport, and I'm certainly not the type, not at all, no - well, I wouldn't quite say extravert, but the more lively you are personally, in speaking or dealing actively with adults, or constantly - having new ideas or even making the odd criticism of seminary instructors, but in a witty or jocular way, more a "master-of-ceremonies" type; they are a great success, I believe.

Q: Yes, hmm (a little insecure).

I: But that of course is a question of mentality. How can that sort of thing be assessed (laughs) or made into a yardstick?

Case D:

I: That all depends on what expectations one has about school life. I had very very low, well, slight expectations on let's say an ideological or educational level as far as my seminary training was concerned. I had no intentions of following a certain, a certain pattern, let's say, or of putting certain ideas into practice through my teaching. I was solely interested in whether the children would like me and that I would do the job that I had to do as well as I could do it. And then try to teach them something personal: those two components, in other words: subject-biased/personal.

Q: And did that work?

I: No! (laughs). It didn't work. I mean, let's put it this way: these pragmatic demands, expectations, they're in any case a bit, they're rather petty, unimportant, not even they worked. And the reasons were a) **because one has had no experience**, one just stands there in front of the class as a human being, not as a teacher. And that is not accepted. Then, secondly, conditions at the seminary school. The children in a seminary school like that always say, aha, here comes another new teacher trainee. One could be the third trainee within a single year, that's something you shouldn't forget, especially what the children all have to put up with. Another one, they say, here comes another one!

Q: They know that of course.

I: They know exactly that one is not an independent teacher, but just someone who gets a lot of stick from above. That's the second point. And then the third point is that the, the pressure from seminary instructors... **That you, they make you feel so small, everything - every word, every gesture, everything. Whoever you are, they'll first destroy you through criticism. All they do is criticize, that was the case with me. And then you are, your self-confidence is zero-level** and then you're supposed to go in front of a class and exude self-confidence and, and knowledge and authority and leadership. You just can't master a conflict like that.

Q: Well how did you solve that problem for yourself?

I: I did it like this. Every time when, when, well, the first and third periods of the postgrad phase were totally chaotic. In the branch school it was better. We were relatively independent there. And for me personally, well it really finished me off, but it was the same with all of us. So it was, well we finished our seminary training feeling that big: very very very small.

Q: And the main factor was the pressure put on you by the seminary instructors?

I: Yes, and the criticism you were exposed to. Towards the end

now, you thought, well, **you'd taught classes for a whole year quite satisfactorily. Or sometimes you really did have a good relationship, got on well with the class.** Then out you came and sure enough, every lesson you held was pulled apart until nothing was left of it. And you thought, my God, what am I? Your self-confidence..that all you'd done the whole year was apparently nothing but rubbish, that nothing you'd ever done was correct. That's the feeling you have.

Q: And all the other problems, like not getting on with classes, followed on from that?

I: **But we did get on with the class,** as we now had a bit more teaching experience, I mean we know on the basis of a year's experience at the branch schools, there we did...we know now how children are taught things. What really got to us was the fact that that wasn't accepted by the seminary instructors.

Q: I mean particularly right at the beginning as well, when you're starting fresh.

I: That was completely...when I think back to...

Q: That was complete chaos.

I: Yes, complete chaos. Inwardly, personally and then this shock at the seminary instructors and the classroom situation. When we were left alone the first time, without a seminary instructor sitting at the back, they went mad, all hell was let loose and (laughs) that was the first shock, how to go about imposing your will on the class for the first time. That's the first step. The second step is that when you're standing there in front of the class and you can keep them all quite, that you can actually convey something, teach something to them. And at the beginning we never managed that at all, because a certain method is necessary for that and that is what you must first learn.

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